Psychological Mechanisms for Eliciting Preferences and Beliefs*

Evan Friedman† Suanna Oh‡ Duncan Webb§

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Abstract

Accurately measuring preferences and beliefs in surveys is crucial for social science research, but standard monetary incentives cannot be used when responses cannot be verified. We study two psychological mechanisms for improving answer quality that can be applied to unverifiable questions: (i) an unexpected bonus payment designed to trigger reciprocity towards the researcher, and (ii) telling respondents that they will later be paid to accurately restate their previously-given answers, which could motivate careful initial answers that are naturally easier to reconstruct. In a large online experiment (N=2,428), the bonus method improves both answer correctness and consistency, and does so more effectively than dropping participants who fail an attention check. This effect is driven by increased effort and reciprocity. The restatement method, however, does not consistently improve answer quality, primarily because participants exert effort trying to memorize their answers instead of answering carefully. These results demonstrate the potential and limitations of using psychological mechanisms to improve the quality of survey responses.

Keywords: restatement method; bonus method; incentives; survey

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†Paris School of Economics and CESifo, evan.friedman@psemail.eu.

‡Paris School of Economics, BREAD, CEPR, and CESifo. suanna.oh@psemail.eu

§Princeton University, dmbwebb@gmail.com

1 Introduction

Social scientists face a fundamental challenge: accurately measuring preferences and beliefs. These measurements matter for testing theories, identifying structural relationships, and for pollsters measuring public opinion. However, survey data quality depends critically on how much effort respondents exert. A typical study can expect 8-12% of respondents to answer carelessly or inattentively (Curran, 2016), and even a low proportion of these respondents can severely degrade data quality by introducing noise and systematic measurement error (Meade and Craig, 2012; Maniaci and Rogge, 2014; Meyer et al., 2015). This problem has become particularly acute with the rise of online surveys and experiments, where there are widespread concerns about data quality (Chmielewski and Kucker, 2020).

Incentivizing answer quality in survey responses takes the form of a principal-agent problem: the researcher (principal) wants to induce respondents (agents) to exert sufficient effort. For some questions, which we call *verifiable*, this is relatively straightforward with standard incentive-compatible designs. We can pay people to get the right answer for questions with objectively correct answers, incentivize belief elicitation, or implement respondents' chosen options for simple preference questions of the form "which do you prefer?".[1] However, many of the most important survey questions in social science are not fully verifiable. They may concern distant future events, subjective feelings, policy preferences, or hypothetical choices, where traditional incentives cannot be applied. Moreover, these are areas where participants may not immediately know the answer to a question without careful thought (Falk et al., 2021a). This raises a fundamental challenge: how can we motivate respondents to think carefully and report truthfully when we cannot verify their answers?

We introduce two methods, based on psychological principles, to improve the quality of survey responses for both verifiable and unverifiable questions. Our first approach, the

[1] See Schotter and Trevino (2014), Schlag et al. (2015), and Barseghyan et al. (2018) for excellent review articles on eliciting beliefs and risk preferences.

Bonus method, provides respondents with an unexpected payment midway through the survey. Building on extensive evidence that reciprocity powerfully motivates behavior (Fehr and Gachter, 2000; Fehr et al., 2002), we hypothesize that this surprise payment acts as a gift that triggers reciprocal effort. While researchers have documented such gift exchange effects in labor markets (Akerlof, 1982; Gneezy and List, 2006; Falk, 2007), its effectiveness in improving survey responses remains unknown. The bonus could help if it motivates respondents to reciprocate, but it could fail if respondents view researchers as too socially distant, if it crowds out intrinsic motivation, or if increased effort remains too psychologically costly (DellaVigna et al., 2022).

In our second approach, the *Restatement* method, we tell participants that they will have to restate their answers to a random subset of questions later in the survey, and will be paid if they restate the same answers they gave previously. This approach exploits a key insight: for questions about stable preferences, beliefs, long-term memories, or reproducible cognitive processes, careful initial responses make it easier for participants to restate later *without* requiring them to memorize their answers. For instance, respondents need not memorize their favorite book or the answer to 21+35 to reproduce these answers later. By contrast, a respondent who answered carelessly or pseudo-randomly would not easily be able to select the same answer again later. Respondents who understand this connection between effort and payments should therefore provide higher-quality initial answers. However, the method could backfire and harm answer quality if participants resort to simply memorizing answers or if they "game the system," e.g., by choosing memorizable but incorrect answers or searching for answers online. Success therefore depends on whether the method encourages effortful thinking or whether it primarily induces memorization or gaming.[2]

[2]One might also be concerned about the efficacy of the *Bonus* or *Restatement* methods on questions that evoke other types of response biases. We do not expect the methods to reduce certain types of biases such as social desirability or experimenter demand effects, as we will discuss in Section 2. The primary goal of this paper is to test whether these methods can increase effort and reduce inattention in surveys, and hence we avoid question types prone to such biases.

We test these methods using a large online experiment (N=2,428) on Prolific. Midway through the survey, participants are randomly assigned to one of five conditions. Our two main treatments test the novel incentive structures: (i) *Bonus* participants receive an unexpected extra payment mid-survey that is explicitly linked to an encouragement to answer questions carefully; and (ii) *Restatement* participants are told they can earn additional payments for accurately restating their answers to 3 randomly selected questions later in the survey. We compare these to two benchmark treatments: (iii) *Conventional Incentives*, in which participants receive additional payments for objectively correct answers; and (iv) *No Incentives*, a control group in which participants receive no additional payments during the survey but receive a larger completion fee. We also evaluate a variant called (v) *Restatement-Points*, which informs participants they will receive non-monetary points for correctly restating their answers, to evaluate whether the restatement method can work at no cost to the researcher. Importantly, the incentives are calibrated so that the overall average payment to participants is approximately the same across groups.

We measure answer quality in three ways. First, we assess (i) *correctness* for verifiable questions with objectively correct answers, including Raven's matrices, simple calculations, and choices between lotteries with a stochastically dominant option. Then, to address the fundamental challenge of measuring response quality for questions where the researcher cannot verify the correct response, we consider two additional measures for unverifiable questions. We measure (ii) internal *consistency*, capturing whether respondents give compatible answers to pairs of differently-worded versions (near-duplicates) of the same question.[3] This follows a literature on test-retest reliability in clinical measurement (Bland and Altman, 1986), but uses it to assess respondent reliability rather than question reliability. Finally, we

[3]For example, we ask "What proportion of total income in the U.S. is earned by the **20% of people who earn the most** (before tax and redistribution)?", and "What proportion of total income in the U.S. is earned by the **80% of people who earn the least** (before tax and redistribution)?" Consistent answers add up to 100%.

measure (iii) *correlations* between closely related (non-duplicate) questions in a supplementary experiment. Respondents are shown pairs of questions from the Pew American Trends Panel Survey (Pew Research Center, 2024) that measure a participant's beliefs and policy preferences about a topic.[4] Answers to questions in a pair are highly correlated in both the original Pew data and our data. More thoughtful responses should strengthen these natural correlations by reducing measurement error.

The *Bonus* method yields modest improvements in response quality. It improves consistency by approximately 3 p.p. (5%), and in some specifications improves correctness by 2 p.p. (3%), although in others it has null effects. Even though the *Bonus* method does not pay more than the *No Incentives* condition, the *Bonus* method's paying of participants midway through the survey and explicitly asking for increased answer quality improves response quality. The improvement in consistency is significantly larger (p=0.10) than the effect of dropping participants who fail an attention check, though the improvement in correctness is not significant (p=0.29). Participants appear to exert more effort in the survey, as indicated by longer response times, increased ability to restate their answers, and self-reports of greater attention and effort. As hypothesized, reciprocity appears to be partially responsible for this effect: a reciprocity index, composed of participants' reports of how thankful they feel for their payments and their willingness to answer more questions for the researcher, increases by 0.16 SD. The effects of the gift-framed payment are larger on easier questions, possibly because these questions are more responsive to changes in participant effort.[5]

The *Restatement* method—telling participants they will be rewarded for restating their

[4]For example, we measure the correlation between a belief (i) "How much of a threat, if at all, is climate change to our country?" and a policy preference (ii) "How important is the following when it comes to being a good member of society?: *Making choices that help reduce the effects of global climate change*"

[5]The modest effects suggest that answer quality is relatively unresponsive to effort. Tellingly, even the *Conventional Incentives* treatment did not lead to a significant increase in correctness, despite direct payments for correct answers (although we cannot reject that the *Bonus* and *Conventional Incentives* have the same effect (p=0.61)). These results extend a growing literature that finds monetary incentives for correctness do not affect response quality, or may even harm it (Mason and Watts, 2009; Danz et al., 2022; Enke et al., 2023).

own previously-given answers—substantially increases internal consistency by 8.6 p.p. (14%). However, the method has little to no effect on correctness—if anything, there is a small negative impact, despite increasing proxy measures of effort (response time and self-reported effort). To understand these results, we consider a multitasking principal-agent framework in which respondents can exert effort along two dimensions: they can either (i) answer carefully, improving both restatement ability and initial answer quality, or they can (ii) memorize answers or "game," improving restatement ability but harming initial answer quality. Post-survey responses indicate that both channels are important: participants claiming to have memorized their answers show significantly lower correctness,[6] while those who report responding more carefully show higher correctness, resulting in a null effect overall. This heterogeneity explains why correctness is not improving on average, but it is nevertheless consistent with improvements in response quality among a subset of participants, suggesting that refined versions of this approach could generate overall improvements.

The contrast between the restatement method's positive impact on consistency and its null effect on correctness presents a puzzle: Does this method genuinely enhance answer quality? Or does it merely encourage recognition of near-duplicate questions? In the latter case, participants may simply be choosing to reproduce their prior answers without answering more carefully initially, in which case consistency does not reflect higher answer quality.

To distinguish between these two explanations, our supplementary experiment (N=1,408) tests whether the restatement method improves initial response quality in the absence of near-duplicate questions. Based on our third measure of answer quality for unverifiable questions—*correlation* between a participants' beliefs and policy preferences about a topic—we find that the restatement method has no effect on average. We interpret our overall set

[6]Gaming strategies are not empirically important: participants' reports suggest they impact less than 5% of questions, and dropping participants who report gaming does not affect the main results. As discussed in Sections 2 and 3, we avoided including questions with salient answer options or Likert scales, which we suspect mitigated the use of gaming strategies.

of results as indicating that the increased consistency effects in the *Restatement* treatment are largely driven by participants recognizing near-duplicate questions and trying to give consistent responses, rather than improving answer quality.

Our findings demonstrate that framing mid-survey payments as unexpected gifts can modestly improve data quality, even for unverifiable questions and particularly for easier questions. Our study also reveals important challenges in using psychological mechanisms to enhance unverifiable survey responses: incentives can lead to multitasking concerns, whereby participants focus on other dimensions of effort (like memorizing their answers) rather than improving underlying response quality.

Related Literature. Our primary contribution is to expand the nascent toolkit for incentivizing high-quality responses to unverifiable survey questions. One strand of prior work, which includes the Bayesian Truth Serum and its variants (Prelec, 2004; Witkowski and Parkes, 2012; Radanovic and Faltings, 2013; Baillon, 2017), peer prediction (Miller et al., 2005), Bayesian markets (Baillon, 2017), choice matching (Cvitanic et al., 2019), and peer betting (Baillon et al., 2025), models respondents as players in a game and establishes truth-telling as a Bayesian Nash equilibrium. These methods vary in the extent to which respondents are assumed to share a common prior, what information is known to the researcher, whether the method can be applied to arbitrary questions or only binary ones, and whether additional questions—such as predictions of others' choices—must be asked.[7] Such approaches can be effective (e.g., see Baillon et al., 2022), but they all require that agents are strategically sophisticated and Bayesian for incentive compatibility, assumptions that may not hold in practice.[8] By contrast, we introduce two simpler mechanisms, rooted in psychological principles, that do not require any strategic or Bayesian reasoning. The instructions are simple to convey in natural language and only negligibly increase the length of

[7] Baillon et al. (2025) provides additional discussion of these methods.

[8] Benjamin (2019) reviews the literature on failures of Bayesian updating.

the survey – both important considerations for "behavioral incentive compatiability."[9] More in line with our psychological approach, Jacquemet et al. (2019) show that people perform better in discrete choice tasks if they first sign an oath to "faithfully and conscientiously fulfill the tasks."

We also contribute to the broader literature on improving survey data quality (Stantcheva, 2023) by developing techniques that directly enhance participant attentiveness rather than filtering out inattentive responses. Current best practices rely heavily on attention checks (Huang et al., 2012; Berinsky et al., 2014; Curran, 2016; Berinsky et al., 2016; Abbey and Meloy, 2017), but these methods risk annoying respondents, increasing attrition, and creating selection bias when used to exclude respondents. Despite not requiring any additional payments, we find our *Bonus* method improves answer quality more than dropping participants who fail an attention check. Other "ex-post" corrections involve collecting multiple measures of the same quantity. For instance, this can be used to reduce classical measurement error (Gillen et al., 2019) or identify participants with greater "self-knowledge," which, when accounted for, can improve the explanatory power of regression models (Falk et al., 2021b). However, these methods require doubling the number of questions in the survey.

Finally, our paper builds on important work in psychology and behavioral science, applying existing insights to survey design. Our bonus method is based on the well-known "gift exchange" effect (e.g. Fehr and Gachter, 2000), whereby subjects reciprocate a "gift" by putting in more effort. The restatement method builds on a literature that explores the relationship between truth-telling and memory, particularly work on "judgements of learning" that researches people's own predictions of their ability to recall (Benjamin et al., 1998; Vieira and Lane, 2012; Besken, 2018; Zimmermann, 2020). Our novelty is to evaluate whether participants' anticipation of their ability to restate their own answers can be used

[9] Danz et al. (2022) show that when the details of the binarized scoring rule (Hossain and Okui, 2013) are made salient, participants may give *less* accurate answers on very simple questions, suggesting that participants may be confused by seemingly artificial incentive schemes.

to improve survey effort.

2 Conceptual Framework: Mechanisms & Limitations

In this section, we provide a simple conceptual framework to illustrate how the bonus and restatement methods may affect the quality, or accuracy, of survey responses (i.e., the probability of answering correctly or truthfully).

In the *Bonus* treatment, subjects receive an unexpected monetary bonus $B > 0$, framed as a gift in exchange for increased effort. While this payment does not depend on accuracy, we hypothesize it increases intrinsic motivation to answer correctly.

In the *Restatement* treatment, respondents learn that they will be paid $R > 0$ for correctly restating their previously-given answers later in the survey. This exploits a simple logic: careful initial responses may naturally be easier to reconstruct later, while careless answers may require significant memorization effort to reproduce. Thus, for questions with reconstructable answers – e.g., concerning stable underlying preferences, long-term memories, or calculations – answering truthfully can be an easier route to correctly restating an answer.

However, the restatement method will improve answer quality only if participants exert effort on answering accurately rather than memorizing answers or gaming the system. To highlight this tradeoff, we suppose that participants can exert effort along two dimensions: by *answering more carefully* in their initial responses ($e_1 \geq 0$) and by *memorizing or gaming* ($e_2 \geq 0$).

We illustrate how the *Bonus* and *Restatement* treatments may affect behavior relative to *No Incentives* using a common framework. Suppose the respondent has the following utility

function:

$$U(e_1, e_2; B, R) = \underbrace{\gamma(B) p_Q(e_1, e_2)}_{\text{intrinsic payoff for accuracy}} + \underbrace{R p_R(e_1, e_2)}_{\text{restatement payoff}} - \underbrace{C(e_1, e_2)}_{\text{effort cost}}, \tag{1}$$

where $p_Q(e_1, e_2)$ is the probability of answering a given question accurately, which depends on both dimensions of effort; $\gamma(B) \geq 0$ is the intrinsic utility from answering accurately, which may depend on the bonus payment B that is paid regardless of accuracy;[10] $p_R(e_1, e_2)$ is the probability of correctly restating the initial answer, which also depends on both dimensions of effort; R is the restatement payment; and $C(e_1, e_2)$ is the cost of effort. $B > 0$ and $R = 0$ corresponds to the *Bonus* treatment, $B = 0$ and $R > 0$ corresponds to the *Restatement* treatment, and $B = R = 0$ corresponds to the *No Incentives* treatment.

For simplicity, we assume linear functional forms for the probabilities $p_Q(\cdot)$ and $p_R(\cdot)$. Answering carefully improves initial accuracy, but memorizing or gaming may hurt accuracy, so $p_Q(e_1, e_2) = p_0 + \alpha_1 e_1 - \alpha_2 e_2$ with $\alpha_1 > 0$, $\alpha_2 \geq 0$. By contrast, answering carefully, memorizing, and gaming all increase the probability of correctly restating, so $p_R(e_1, e_2) = p_0' + \beta_1 e_1 + \beta_2 e_2$ with $\beta_1 > 0, \beta_2 > 0$. We assume that costs are separable and convex, i.e., $C(e_1, e_2) = c_1(e_1) + c_2(e_2)$ with $c_j(0) = 0$, $c_j'(e_j) > 0$, and $c_j''(e_j) > 0$ for $e_j > 0$ and $j \in \{1, 2\}$.[11]

Bonus treatment. When $R = 0$, the participant will set $e_2 = 0$ and e_1 such that $\gamma(B)\alpha_1 = c_1'(e_1)$. Increasing the bonus payment will therefore increase effort e_1 and initial answer accuracy p_Q if and only if $\gamma'(B) > 0$. In other words, if the gift increases the intrinsic payoff for accuracy (as per the evidence on reciprocity in psychology, e.g., Fehr and Gachter, 2000), then accuracy will increase.

[10] We allow for $\gamma(0) > 0$, so there can be an intrinsic desire to answer accurately even without any bonus payment.

[11] We also assume two technical conditions: that each $c_j(\cdot)$ and $\gamma(\cdot)$ are twice continuously differentiable, so that the previous conditions are well-defined; and that $C(\cdot)$ and the other parameters ensure that the optimal effort choices are strictly interior as long as $\gamma(B) > 0$ and $R > 0$.

Restatement treatment. We also derive the conditions under which increasing the restatement incentive R can increase initial accuracy p_Q.

Proposition 1.

$$\frac{\partial p_Q^*}{\partial R} > 0 \quad \textit{if and only if} \quad \frac{\alpha_1 \beta_1}{c_1''(e_1^*(R))} > \frac{\alpha_2 \beta_2}{c_2''(e_2^*(R))} \tag{2}$$

Proof. See Appendix B. □

Increasing the restatement payment R will thus tend to improve accuracy if (i) answering carefully contributes strongly to answering accurately (high α_1); (ii) answering carefully contributes strongly to one's ability to restate in the next round (high β_1); and (iii) exerting more effort on e_1 is relatively "cheap" (low $c_1''(.)$). Conversely, increasing R will tend to worsen accuracy if (iv) memorizing or gaming strongly worsens accuracy (high α_2); (v) memorizing or gaming is effective at improving restatement (high β_2); and (vi) memorizing or gaming is "cheap" (low $c_2''(.)$).

Limitations and scope. We do not expect the *Restatement* or *Bonus* methods to reduce social desirability bias or experimenter demand effects. Indeed, for questions that evoke such concerns, these effects may be reinforced, leading to a reduction in accuracy. As our framework highlights, the *Restatement* method may also backfire whenever there is any particularly salient answer. Because the salient answer is easy to remember, choosing it is a low-cost gaming strategy that participants may pursue instead of simply answering more carefully. In our experiment, we therefore avoided including questions that may evoke social desirability or experimental demand concerns, as well as questions with salient answers. In particular, we avoided ordered Likert scale questions where Restatement may induce gaming by selecting salient (first or last) answers. As detailed in the next section, our questions are mostly multiple choice, and we took measures to mitigate the attractiveness of gaming strategies.

3 Experimental design

The experiment aims to test whether the *Bonus* and *Restatement* methods improve answer quality compared to conventional incentive approaches.

The main experiment consisted of five between-subject treatments, as detailed in Section 3.2 below. Each treatment followed the same structure, consisting of five stages: *Background*, *Main*, *Spillover*, *Restatement*, and *Debrief*. The primary treatment variation occurs at the start of the main stage, when participants are given different information about how they will be paid (corresponding to each of the five treatments). The primary outcome measures—correctness and consistency—are based on the questions in this stage. Figure A.1 summarizes the overall structure of the experiments.[12]

Recruitment. We recruited US-based adults on Prolific, an online survey platform, between January and February 2023. We selected Prolific because it is known to have high data quality relative to similar platforms (Peer et al., 2022), suggesting that our effects are likely to be lower bounds relative to other platforms like Amazon MTurk. All payments were in British pounds (£), with £1 ≈ $1.25. The survey took 38.6 minutes to complete on average.

Pre-registration. The study was preregistered on AsPredicted (main experiment and supplementary experiment). The only deviations are the inclusion of LASSO controls in some specifications, discussed and justified below (Section 3.4), the pooling of verifiable and unverifiable questions in the main specification (the unpooled results, reported in Table A.12, are very similar).

[12]The full questionnaires are available here and here.

3.1 Survey sections

Stage 1: Background. The background stage had 30 unincentivized questions about standard demographics and household expenditures. It was designed to create realistic survey conditions before introducing our experimental treatments. We hypothesized that survey fatigue would enhance the effectiveness of our novel incentives, so this lengthy initial stage deliberately tired participants while masking the study's focus. An attention check question also verified participant engagement, but we did not drop those who failed it.

Stage 2: Main. At the start of this stage, participants received the instructions relevant to their assigned treatment, which varied the payment scheme they followed (see Section 3.2). Participants then answered 14 *correctness* questions and 8 *consistency* questions, largely taken from other studies or surveys to ensure relevance to practitioners. The questions in this stage were randomly ordered at the participant level.

Correctness questions (detailed in Table A.1). The correctness questions were designed to have objectively correct answers and to be plausibly responsive to effort (rather than simply being general knowledge questions that one was likely to either know or not know). They included simple mathematical calculations, Raven's matrices, items from the cognitive reflection test (Frederick, 2005), dot-counting exercises (Dewan and Neligh, 2020), reading comprehension exercises, and choices from lotteries with a first-order stochastically dominant option.[13]

Consistency questions (detailed in Table A.2). The consistency questions were 4 pairs of reworded or obfuscated near-duplicate questions, which allow us to test whether participants gave internally consistent answers within a pair. The questions concerned participants' beliefs about inequality, education, and immigration, mimicking common styles of survey questions (e.g., see Alesina et al. (2018)). For example, participants were asked their beliefs

[13]We also included 2 exploratory questions testing Bayesian updating that we preregistered to be excluded from the main analysis, given the evidence suggesting that even very strong incentives cannot improve Bayesian updating (Enke et al., 2023). The questions are shown in the notes to Table A.1.

about what proportion of income in the U.S. would go to the top 20% of earners in question (3a), and to the bottom 80% of earners in question (3b). Their answers were deemed to be internally consistent if they summed to 100%. The idea is that if preferences or beliefs are stable during the survey, then any variation across multiple elicitations is likely driven by measurement error. Greater consistency thus implies greater answer quality.[14] We used near-duplicates instead of exact duplicates to reduce the probability that the restatement method would improve consistency purely through memorization rather than through increased attentiveness. A key advantage of the consistency measure is that it can be applied even to unverifiable questions whose answers cannot be checked by the researcher (e.g., because they have no objective answer).

Verifiability of consistency questions. Our incentive methods may have different effects on verifiable and unverifiable questions. Participants might reciprocate the bonus payment less if they know their answers cannot be checked, and the restatement method might encourage consistency at the expense of accuracy for such unverifiable questions. To test these possibilities, we randomly assigned participants to answer either verifiable or unverifiable versions of our consistency questions.[15] For instance, some participants estimated the income share of the top 20% of U.S. earners **in 2050**—an unverifiable prediction—while others estimated it **in 2023**, which has a known correct answer (see question (3a) in Table A.2).

Question and response order. Most questions were multiple choice, and we told participants that response options would appear in random order. This discouraged participants in the *Restatement* arm from gaming the system by following simple heuristics, e.g., "always pick the first option". We find no evidence that the treatments increased the probability of

[14]This follows work on test-retest reliability in clinical measurement (Bland and Altman, 1986) and is related to the practice in economics of using multiple measurements of the same quantity to reduce measurement error (Gillen et al., 2019).

[15]Only verifiable questions were used for *Conventional Incentives* so that we could always pay participants for objectively correct answers. We also only included verifiable questions for *Restatement-Points* in order to reduce survey costs. The sample size for these treatments is therefore approximately 300 instead of 600. Question verifiability had little effect on the respondents' behavior (see Table A.12).

picking the first or last option (Table A.17). Questions in the main stage were also randomly ordered to avoid systematic order effects.

Stage 3: Spillover. To test whether the effects of incentives in the main stage persist (or "spill over") once the incentives are removed, we include 10 additional questions that were explicitly unincentivized. These questions included 3 correctness questions and 1 pair of consistency questions. Both positive and negative spillovers were a priori plausible: the increased attention in the main stage could tire out participants, or it could habituate participants to be more attentive throughout the whole survey.

Stage 4: Restatement. In all treatments, we randomly selected 3 questions from the main stage at the participant level, and asked participants to restate the same answers they gave earlier. For the *Restatement* and *Restatement-Points* treatments, participants were informed about the restatement stage at the start of the main stage to see how it affected their behavior in the main stage. For other treatments, the restatement stage came as a surprise. *Restatement* participants were paid £1 per correctly restated answer, while *Restatement-Points* participants were only awarded non-monetary points. Other treatments were either paid £0.10 or £1.[16]

Stage 5: Debrief. The final stage includes questions designed to understand the mechanisms behind how the incentives affected survey behavior. This includes questions about reciprocity, attention, and effort.

[16]We paid £0.10 per correctly restated answer in the *Conventional Incentives* and *Bonus* treatments to ensure the average total payments were similar across groups. This allows us to more readily compare across groups elicited measures of reciprocity (toward the researchers) from the debrief stage that followed. *No Incentives* participants were randomized to either receive £0.10 or £1 per correctly restated answer. The £0.10 payments keep total payments similar across arms, following similar logic. The £1 group allows us to compare restatement performance between *No Incentives* and *Restatement*, i.e., when the restatement stage comes as a surprise versus when it is known in advance, holding fixed the restatement payments.

3.2 Treatments

Participants were randomly allocated into one of five treatment arms. Starting at the main stage, instructions varied by treatment (see Appendix C for full scripts):

1. **Bonus** (N=613). Participants were told that they would receive an "extra bonus" payment of £3 (in addition to the completion fee).

2. **Restatement** (N=599). Participants were told about the restatement stage, i.e., that they would receive £1 for correctly restating their answers for each of 3 randomly selected questions from the main stage.

3. **Restatement-Points** (N=300). Participants were also told about the restatement stage, but were told they would receive 1 point per correctly restated answer. We explicitly state that these points have no monetary value.

4. **Conventional Incentives** (N=307). Participants were told they would be paid based on their answers to 3 randomly selected questions from the main stage. If the question had an objectively correct answer, they would receive £1 for giving the correct answer. If the question asked respondents for their preference (e.g., over lotteries), they would receive their chosen alternative.

5. **No Incentives (Control)** (N=609). Participants were not given additional information, but were simply asked to pay attention and put effort into answering the questions.

The language describing the incentive schemes was kept as similar as possible across treatments. To reduce any differential impact of experimenter demand, all instructions included the phrase: "we would like you to pay attention and put effort into answering the questions in this section". We included multiple comprehension checks at this stage to make sure participants understood their incentive scheme.

Completion fee. All participants were recruited with an advertised completion fee of £4. Prior to giving consent to participate, they were told again that they would receive £4, except those in the *No Incentives* and *Restatement-Points* treatments, who were told that they would receive £7.[17] The completion fee was higher in these treatments so that the *total payment* was approximately the same across all arms.[18] This provides a conservative test of the effectiveness of *Restatement* and *Bonus*, where only the method of payment differs rather than the amount.

3.3 Outcomes and specification

Our primary outcomes are two measures of answer quality—correctness and consistency. These are based, respectively, on the correctness and consistency questions from the main stage (see Section 3.1).

Correctness. Correctness is an indicator for whether respondent i gave the objectively correct answer to correctness question j.

Consistency. Consistency is an indicator of whether respondent i gave the same answers across a pair of near-duplicate consistency questions j. Unlike correctness, consistency does not require verifiability, and so we use it to measure answer quality in both verifiable and unverifiable treatments.

Consistency and correctness are positively correlated: participants who are more correct also tend to be more consistent across all treatment groups (Appendix Table A.3). This suggests that they are both capturing an important aspect of overall answer quality.

Empirical specification. Following our preregistered analysis, we run an OLS regres-

[17] To ensure that participants were aware of the completion fee, in all treatments, participants had to answer a question acknowledging the amount of the completion fee. The language was identical across the treatments apart from this amount. In particular, we did not emphasize when the completion fee was more than advertised.

[18] Mean total payments (including restatement payments) were: £7.27 for *Bonus*, £7.26 for *No Incentives* (those who received £0.10 per correctly restated answer), £7.11 for *Conventional Incentives*, £6.70 for *Restatement*, and £7 for *Restatement-Points*.

sion of the form:

$$Y_{ij} = \beta_1 Bonus_i + \beta_2 Restate_i + \beta_3 RestatePoints_i + \beta_4 Conventional_i + \mathbf{X}_i'\Gamma + \mu_j + \varepsilon_{ij} \quad (3)$$

where i is the participant, and j is the question (for correctness questions) or question-pair (for consistency questions). $Y_{ij} = 1$ if the respondent answers correctly or if their answers in a pair are consistent, and is 0 otherwise. *No Incentives* is the excluded category. μ_j are question or question-pair fixed effects, and standard errors are clustered at the participant level. In some specifications, we include a vector of controls $\mathbf{X}_i$ selected by double LASSO (Belloni et al., 2014) to account for confounding imbalances (see Appendix D.2 for list of controls selected). We analyse correctness and consistency separately, pooling all questions in each category into a single regression.

Additional outcomes. To understand mechanisms, we also test whether the methods impact self-reported effort and attention, response times, the probability of correctly restating earlier answers, and measures of reciprocity towards the researchers. We describe each outcome in more detail when we present the results.

3.4 Sample characteristics, balance and attrition

Table A.4 shows the demographic characteristics of the main sample across treatments. The sample demographics are similar to the US population: 48% of participants are female, and the sample is on average 40 years old, has $64,000 in household income (slightly below the US average), a 60% college completion rate (similar to US average), but with a low proportion of employed (65%) and a high proportion of white persons in the sample (73%).

Relative to the *No Incentives* group, we find no significant overall imbalances in the *Conventional Incentives*, *Restatement*, or *Restatement-Points* groups (p of a joint F-test, 0.32, 0.39, 0.87 respectively). However, there is evidence of a random but meaningful imbalance

in the characteristics of the *Bonus* group (p of F-test = 0.05). This imbalance appears to generate a downward bias on our estimates, since the *predicted* correctness and consistency scores based purely on demographic characteristics are significantly lower in the *Bonus* arm (Table A.5, columns 3-8).[19] To control for this imbalance, all analyses include double LASSO-selected controls unless otherwise specified (see Appendix D.2).

4 Main results: impact on correctness and consistency

Table 1 and Figure 1 show the effects of the treatments on correctness and consistency.

The *Bonus* method improves consistency and, in our preferred specification with LASSO controls, correctness. Consistency improves by approximately 3 p.p. (5%, p=0.02). Correctness improves by a more modest 2.1 p.p. (2.8%, Table 1, column 2, p=0.03) in the specification with LASSO controls.[20]

Restatement and *Restatement-Points* both have no detectable effect on correctness (columns 1-3 and panel A, p=0.19 and 0.15 respectively), and the coefficient estimates are slightly negative. We are well-powered to rule out even relatively small effects, with an implied minimum detectable effect of ±3.1 p.p. (4.1%) in the preferred LASSO specification.[21] This is thus clear evidence that objective correctness does not improve. By contrast, participants are 8.4 p.p. more likely to give consistent answers in the *Restatement* arm (columns 4-6 and panel B, p<0.001), and 5 to 7 p.p. more likely in the *Restatement-Points*

[19]The imbalance is not driven by differential attrition over the course of the survey, since (i) overall attrition is small, with only 3.2% dropping out of the survey from the start, and only 1.4% dropping out after the demographics section was completed; (ii) the rates of attrition do not differ across treatment groups (Table A.5, columns 1 and 2); (iii) the difference in predicted correctness and consistency scores in the *Bonus* group remains robust even after adjusting for attrition using inverse-probability weights (Table A.5, columns 5 and 8).

[20]Without these controls, the effect is no longer significant (Table 1, column 1, p=0.59), but this reflects sample composition rather than treatment effects. The *Bonus* group started with demographic characteristics that predict 1.4 percentage points fewer correct answers (see Table A.5, columns 3-5). We prefer the LASSO specification because it better accounts for this baseline imbalance.

[21]This assumes α=5%, β=80%, implying that the minimum detectable effect is $2.8\hat{\sigma}$ where $\hat{\sigma}$ is the standard error of the estimate.

Table 1: *Impact on correctness and consistency*

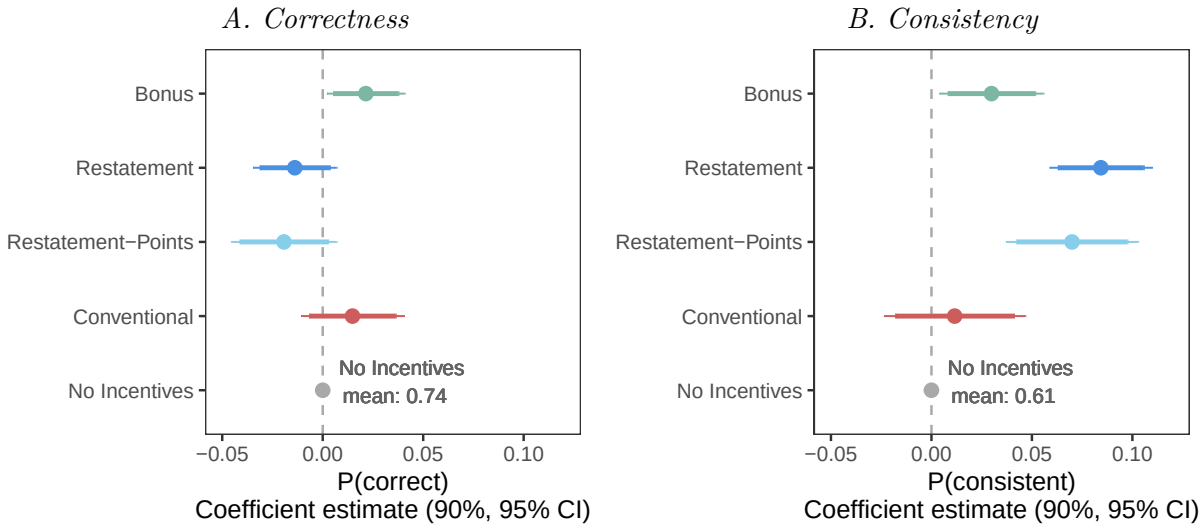
	Correct Answer (=1)		Consistent Answer (=1)	
	(1)	(2)	(3)	(4)
Bonus	0.006	0.021**	0.026*	0.030**
	(0.012)	(0.010)	(0.014)	(0.013)
Restatement	−0.020	−0.014	0.086***	0.084***
	(0.013)	(0.011)	(0.014)	(0.013)
Restatement-Points	−0.020	−0.019	0.054***	0.070***
	(0.016)	(0.013)	(0.017)	(0.017)
Conventional	0.011	0.015	0.000	0.012
	(0.014)	(0.013)	(0.018)	(0.018)
Num. observations	33 992	33 992	9712	9712
Num. participants	2428	2428	2428	2428
Dep. var. control mean	0.74	0.74	0.61	0.61
R-squared	0.07	0.14	0.22	0.25
Question FEs	X	X	X	X
LASSO controls		X		X
p-val: Bonus=Conv.	0.74	0.61	0.14	0.31
p-val: Restate.=Bonus	0.03**	0.00***	0.00***	0.00***
p-val: Restate.=Conv.	0.04**	0.03**	0.00***	0.00***
p-val: Restate.=Restate-Points	0.99	0.69	0.06*	0.38
p-val: Restate. (Mon. + Points pooled)=0	0.07*	0.11	0.00***	0.00***

Notes: Uses specification from Equation 3. Observation is the (participant × question) for columns 1-2, (participant × question-pair) for columns 3-4. Omitted category is the *No Incentives* arm. Outcome is a binary indicator for a correct answer in columns 1-2, and a binary indicator for a consistent pair in columns 3-4. Selected LASSO controls are in Appendix D.2. Standard errors clustered at the participant level.

arm (p<0.001). Below, we discuss evidence that this increase in consistency likely does *not* represent an increase in answer quality, although it may increase answer quality in a subset of respondents. Rather, despite no explicit incentive for consistency, the restatement method leads participants to more easily recognize duplicates, and then repeat their previous answers.

Robustness. The results are robust to (i) adjusting for multiple hypothesis testing (Table A.8); (ii) using randomization inference to calculate p-values (Table A.9, columns 1 and 4); (iii) using an inverse-covariance weighted index of correctness and consistency ques-

Figure 1: *Plotting the estimated treatment effects*



Notes: Panel A is from column 2 of Table 1; panel B is from column 4. Both use LASSO controls and question fixed effects (see Appendix D.2).

tions (Anderson, 2008) (Table A.9, columns 2 and 5);[22] (iv) dropping the 11.1% participants who failed an attention check question in the first stage of the survey (columns 3 and 6). The fact that the improvements driven by *Bonus* persist among those who passed an attention check suggests that answer quality is improving across the distribution of respondents, not only among the most inattentive.[23] The results are also robust to controlling for participants' perception of the purpose of the experiment (Figure A.8, Table A.18), suggesting that the *Bonus* method's effects are not solely driven by experimenter demand effects and are generalizable to settings where participants think that the survey designer has motives other than high-quality responses.

Heterogeneity by question difficulty. The effectiveness of the methods varies across questions. Figure A.2 presents the effects individually for each question or question-pair, along with F-tests of across-question heterogeneity (bottom right panel). The *Bonus* treat-

[22]This index gives more weight to questions that are uncorrelated with other questions and thus provide more independent information about answer quality. It may thus better represent answer quality across a broader set of question types.

[23]We also find no heterogeneity by baseline response time, a proxy of initial attentiveness (Table A.11). This similarly suggests that attentiveness improves across the distribution of respondents.

ment's effects on correctness show significant heterogeneity (p=0.10), and we find systematic evidence of stronger impacts on easier questions: for questions where the the average correctness in *No Incentives* exceeds 74.8% (the median across questions), the *Bonus* increases correct answers by 3.4 p.p. (Table A.10, 4.6%, p=0.01). No such effect exists for harder questions. This is important for practitioners; the *Bonus* method is more likely to be helpful for relatively easy questions. The *Restatement* treatment also results in heterogeneous effects across correctness questions (Figure A.2, p=0.03), but this is not systematically explained by question difficulty (Table A.10).

Heterogeneity by verifiability. Participants who faced *verifiable* consistency questions were slightly less consistent overall (Table A.12, column 1, 2.1 p.p., p=0.05), possibly because participants prioritize accuracy over consistency when their answers can be verified. However, this effect did not vary across treatments (Table A.12, column 2), suggesting that the bonus and restatement methods' efficacy does not depend on the use of unverifiable or verifiable questions.[24]

Spillover. The incentive effects on answer quality persist even when we remove the incentives (Table A.13). In the *spillover* stage, participants answer five additional questions without any monetary incentives: three test correctness, one pair tests consistency, and one is an attention check question. The *Bonus* treatment improves correctness by 2.9 percentage points (column 2, p=0.07), while *Restatement* improves consistency by 7.8 percentage points (column 3, p<0.001). These effects match the magnitude of improvements in the main stage (p of difference: 0.44 and 0.63 for *Bonus* and *Restatement*, respectively). No treatment affects attention check performance (column 1), indicating that the incentives do not reduce participant attentiveness. These results are encouraging for practitioners: rather

[24]The consistency effects are not larger when the near-duplicate question pairs are more "obvious" to respondents. For example, they not affected by whether a pair randomly appears early or late in the survey (Table A.12, column 3), or when near-duplicate questions appear closer together (column 4). This suggests the effects are not driven by participants deliberately aiming for consistency after detecting near-duplicate pairs.

than exhausting participants, the incentives generate positive spillovers that improve answer quality throughout longer surveys.

Comparison to attention checks. The effects of the *Bonus* method are substantially larger than that of the standard technique of dropping participants who fail an attention check. Dropping the 11.1% of participants who fail the attention check only increases correctness by 1.1 p.p. and consistency by 0.7 p.p. (Table A.14). This implies that the *Bonus* method, despite not incurring any additional costs to the experimenter relative to the *No Incentives* group, produces 4.2x larger gains in consistency (p of difference: 0.10), while obtaining similar improvements in correctness (p of difference: 0.29). This means dropping those who fail the attention check produces minimal gains in response quality while reducing the sample size.[25]

5 Mechanisms

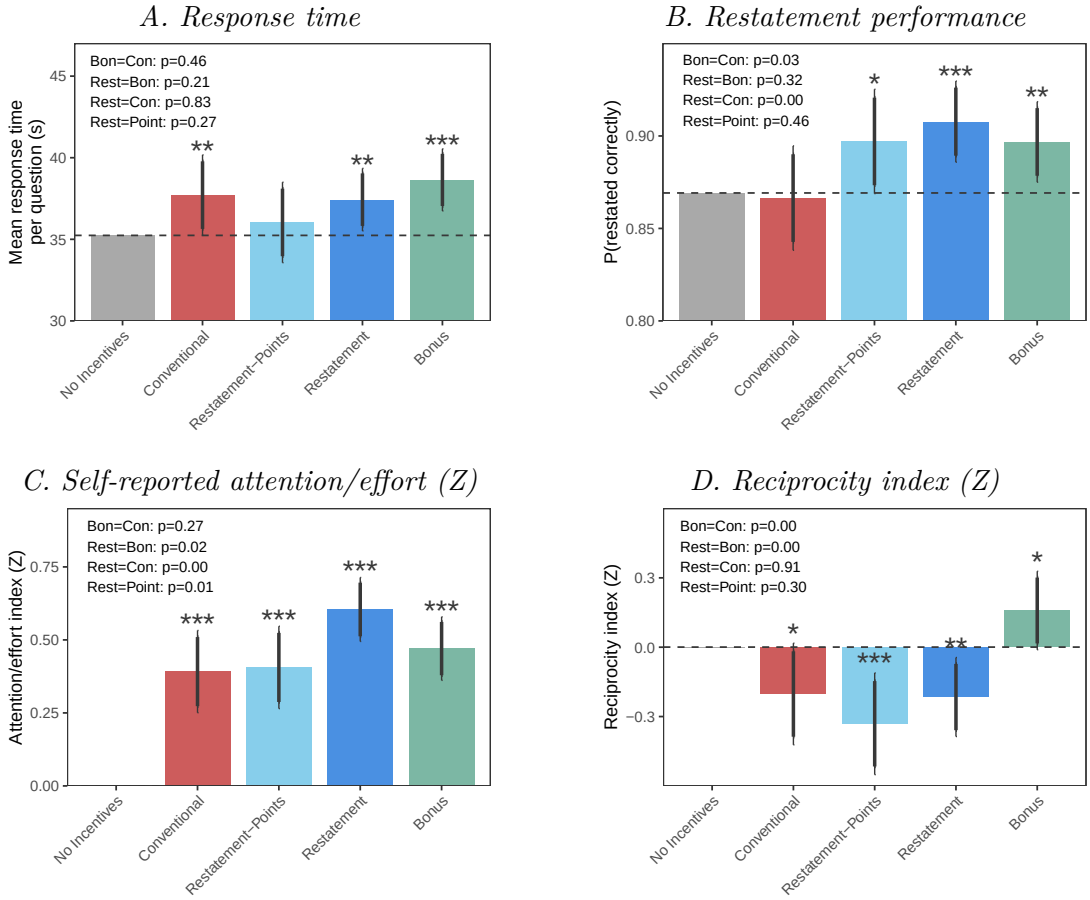
In this section, we explore the mechanisms behind the *Bonus* and *Restatement* incentives. We find that the *Bonus* method improves effort and reciprocity, while the *Restatement* method does not meaningfully improve average answer quality because a substantial portion of respondents focus on memorizing their answers instead of answering carefully.

5.1 Impacts on effort and reciprocity

Response time (Figure 2, panel A). The *Restatement* treatment increases average response time per question by 2.2 seconds (6%, p=0.03) in our preferred specification that controls for baseline response times in the background stage, although this effect is no longer significant

[25]Those who pass the attention check are 6.4 p.p. (p<0.001) more correct and 4.8 p.p. (p=0.001) more consistent than those who fail. These numbers are in line with other work using attention checks, e.g., Alvarez and Li (2021) find that attention checks are failed by 15% of respondents, and those that fail are 6 p.p. more likely to misreport their voter turnout.

Figure 2: *Impact on effort and reciprocity*



Notes: Panel A (*Response time*): outcome is the mean response time for questions in the main stage. Panel B (*Restatement performance*): outcome is the probability of correctly restating for the 3 restatement questions. Panel C (*Self-reported attention/effort*): participants were asked how their incentive scheme affected how much *attention* they paid (from *Greatly decreased* (1) to *Greatly increased* (5)), and a similar question for how much *effort* they exerted; and whether they answered as quickly as possible or as carefully as possible on a 0-5 scale. We normalize each question to have mean of 0 and standard deviation of 1 in the *No incentives* (control) group, then the outcome is the normalized average all 3 questions. Panel D (*Reciprocity index*): participants were asked how thankful they were for their payment (scale 1-5) and whether they were willing to answer three more questions at the end of the survey (yes/no). Outcome is the average of the normalized Z-score of both questions, with mean of 0 and standard deviation of 1 in the control group. All models include all controls selected by double LASSO for both consistency and correctness measures. Thick bars are 90% confidence intervals, thin bars are 95%, calculated relative to control group (*No Incentives*). Stars indicate differences with *No Incentives* (*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$). Panel A (*Response time*) is an ANCOVA specification that controls for baseline response time in the demographic stage.

when removing that control (effect=1.6 seconds, p=0.12). The *Bonus* treatment increases average response time in the main stage by 3.4 seconds (10%, p<0.01 both with and without

baseline control).[26]

This suggests that participants respond to the bonus incentives, and possibly the restatement incentives, by spending more time answering questions, indicating that they are exerting more effort. The effects on response time are similar for both consistency and correctness question types (p-values for differences range from 0.63 to 0.96). This uniform increase in response times suggests participants respond to incentives by increasing their overall effort rather than focusing specifically on either consistency or correctness. In Section 5.2, we explore whether, in the *Restatement* treatment, this effort is focused on improving answer quality or simply memorizing answers.

Restatement performance (Figure 2, panel B). The *Restatement*, *Restatement-Points*, and *Bonus* treatments all increased the fraction of questions correctly restated, and all to a similar degree. These treatments increased this fraction from 87% in *No incentives* to 90-91% (effects between 2.8 and 3.9 p.p.). This improvement likely stems from increased effort during the main stage rather than the restatement stage itself. In the *No Incentives* group, offering £1 versus £0.10 for correct restatements did not affect performance (p=0.13), suggesting that additional payments during the restatement round arrive too late to improve restatement performance. The *Bonus* method's effectiveness at improving restatement performance is particularly revealing: because participants were never told they would need to restate answers later, their improved performance likely stems from greater effort toward answering questions rather than strategic behaviors like memorization.

Self-reported effects on attention, effort, and care (Figure 2, panel C). In the follow-up questions of the debrief stage, participants also reported that the incentives made

[26] As with all online surveys, there are a few instances of very high response times, most likely indicating that participants left their computers for extended periods. Winsorizing these outliers at 1% and 5% levels, for both individual questions and participant-level times, does not significantly change any coefficients. Under the preferred specification, *Restatement-Points* also does not have a significant effect on response time (p=0.53), and we cannot reject that it has the same effect as the monetary *Restatement* method (p=0.27).

them pay more attention and effort. After reminding participants of the incentive scheme they were under, we asked them explicitly how this affected how much *attention* they paid and how much *effort* they exerted in the main stage.[27] We also asked them to rate whether they responded to questions as quickly as possible or as *carefully* as possible on a 0-5 scale. All treatments led to substantial increases in a combined index of these questions of 0.39–0.60 SD (all $p<0.001$).[28]

Responsiveness to effort. Multiple results suggest that answer quality is relatively *unresponsive to effort*, i.e., even when participants try to answer more carefully, answer quality does not improve. For example, even though *Conventional Incentives* increase response time and self-reported attention, participants' correctness does not improve. Participants seem to recognize this: while participants in the *Restatement-Points* arm report increasing attention and effort (see Figure 2, panel C), in the supplementary experiment they do not predict that they will have answered more questions correctly.[29] This awareness of the weak relationship between effort and accuracy (a low α_1 in Section 2) helps explain the modest effects of *Restatement*, consistent with findings that financial incentives only weakly affect accuracy in other settings (Enke et al., 2023). On the other hand, the *Bonus* intervention's success in increasing correctness, even where *Conventional Incentives* failed, suggests it is a particularly promising strategy for improving survey response quality.

[27] E.g., for *Restatement*: "You were told during the survey that you would be paid for correctly restating some of your answers from [*the main stage*]. How did this affect how much **attention** you paid when answering the questions in that section?". For *No Incentives*: "You were told at the beginning of the survey that your payment for completing the survey would be £7 instead of the £4 which was advertised on Prolific. How did this affect how much **attention** you paid when answering the questions in [*the main stage*]?". Answer options ranged from "Greatly decreased attention" (1) to "Greatly increased attention" (5). We also included analogous questions asking about the effects of incentives on **effort**.

[28] While these measures are potentially subject to experimenter demand effects, the *No Incentives* group was also explicitly reminded about their higher completion fee of £7 (as opposed to the advertised rate of £4) when answering these questions, so it is not clear that experimenter demand should be *higher* in the other treatment groups.

[29] We asked participants to estimate the proportion of correct responses they gave for questions in the main stage with an objectively correct answer, between 0% and 100%. Both *No Incentives* and *Restatement-Points* predict a mean correctness rate of 67% (p of difference = 0.60).

Reciprocity (Figure 2, panel D). The *Bonus* treatment increased participants' reciprocity toward researchers. We find an increase of 0.16 SD (p=0.07) in a reciprocity index based on self-reported gratitude for payment and willingness to answer 3 additional questions at the end of the survey. This effect stems primarily from increased gratitude; *Bonus* participants are not willing to answer more questions, suggesting that reciprocity may be primarily channeled to increased answer *quality* in the main stage rather than willingness to increase answer *quantity*. In contrast, the *Restatement* treatments reduced reciprocity, possibly because participants are frustrated by longer survey times (39 minutes in *Restatement* versus 37 minutes in *No Incentives*) or by payments conditional on their behavior, in line with evidence that attention checks can irritate respondents (Stantcheva, 2023).

5.2 Restatement: answering carefully, memorizing, or gaming

Our results show increased effort under the *Restatement* method. As our theoretical framework illustrates, however, this increased effort could either improve or harm answer quality, depending on how participants direct it. Participants facing restatement incentives could respond by answering more carefully in the main stage (improving answer quality), by attempting to memorize their answers (likely harmful or neutral for answer quality), or by gaming the system through strategies like choosing memorable answers or writing down answers (potentially harming answer quality). We examine these channels to understand why the *Restatement* method improves consistency but not correctness. We find that gaming does not seem to be a major concern, but many participants do try to memorize their answers in a way that harms their performance on correctness questions. In contrast, the *Restatement* method does seem to increase correctness for a subset of respondents who answer more carefully.

Gaming. Gaming behavior appears rare and does not explain our findings. In the debrief stage of the *Restatement* treatment, we asked participants if they chose easy-to-

remember answers, searched online, or wrote down answers. The fraction of participants who claimed to engage in these activities for *any* questions was only 14.5%, 4.3%, and 5.8%, respectively (Table A.15).[30] Moreover, participants who followed these strategies reported doing so only for a low proportion of questions, so less than 5% of questions were affected overall. Gaming strategies do not drive the effects of *Restatement*: excluding the 177 participants who reported any gaming does not alter our main findings on consistency and correctness (Table A.16).

Answering carefully and memorizing. Rather than gaming, most participants in the *Restatement* treatment instead focused either on *answering more carefully* or *memorizing answers*, following the two types of effort outlined in the theoretical framework (Section 2). In the debrief stage of the *Restatement* treatment, we asked participants to describe the strategies they used to improve recall performance using open-ended free response, which we coded using OpenAI's API (see details in Appendix Section D.1). For example, one participant wrote, "I read the questions slowly and carefully several times before answering" (coded as careful answering); another wrote, "I tried hard to remember my answers so I could restate them later" (coded as memorizing). The analysis implies that 52.6% of participants reported answering more carefully in the main stage and 25.0% reported trying to memorize their answers (Table A.15).[31]

Heterogeneity by answering carefully vs. memorizing. Performance varied substantially based on participants' reported strategies (Figure 3). Participants who emphasized careful answering achieved both higher correctness (β=4.3 p.p., p<0.001) and consistency (β=10 p.p., p<0.001) compared to the *No Incentives* group. Memorizers, however, per-

[30] In the supplementary experiment, we also asked whether participants used ChatGPT, and only 1.8% of respondents answered yes. *Restatement* participants also do not select first or last options more often in randomly ordered response lists (Table A.17), suggesting they are not prioritizing salient (easy-to-remember) answers.

[31] Only 0.7% of respondents' main strategy was gaming; 32.7% reported no particular strategy; and 0.3% some other strategy.

formed 3.6 p.p. worse on correctness (p=0.04), suggesting that effort spent on memorization may crowd out attention on the questions themselves. These correlational results do not necessarily represent causal effects of the strategies, but they nevertheless align with the theoretical prediction that restatement incentives can improve answer quality if participants answer carefully, and harm it if they attempt to memorize. This heterogeneity can explain the overall null effects on correctness.

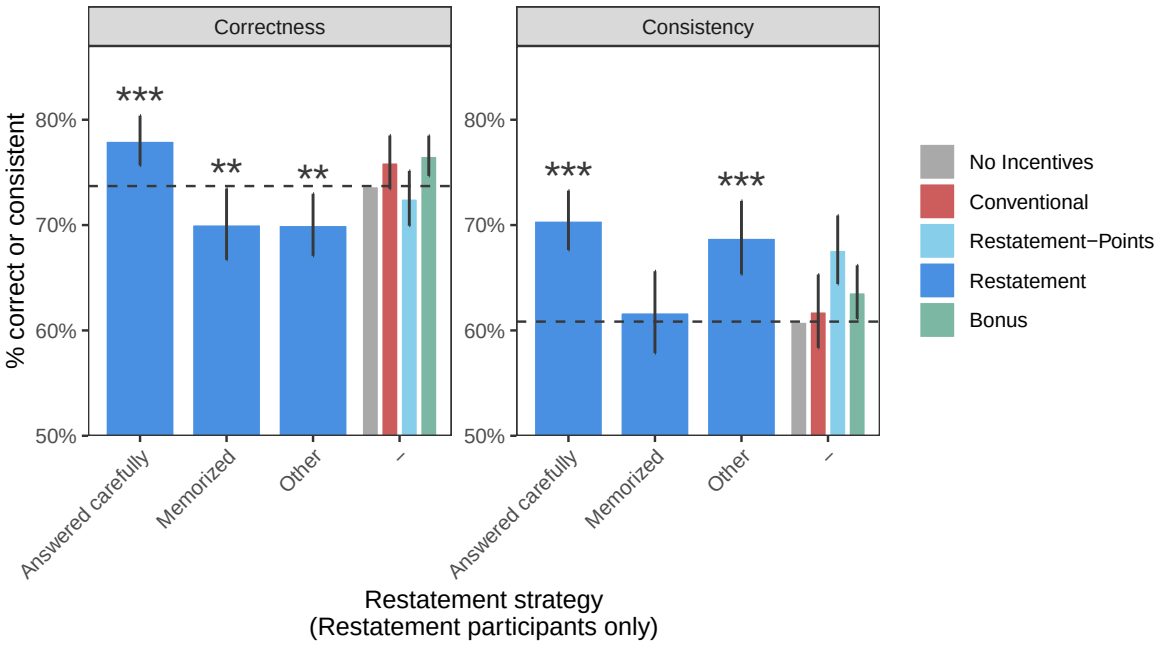
Despite focusing explicitly on remembering answers, memorizers showed no overall improvement in consistency relative to the *No Incentives* group (Figure 3). This null effect masks an important pattern: memorizers achieved slightly higher consistency on simple reworded duplicates (β=3.5 p.p., p=0.06, questions 2 and 4 in Table A.2), but *not* on questions that require careful attention to reversed wording (p=0.60, questions 1 and 3). This selective improvement suggests that memorizers engage in superficial recall rather than careful attention to questions.

In sum, our evidence reveals both channels hypothesized in our theoretical framework: some participants respond to restatement incentives by increasing attention, while others focus on memorization in ways that harm answer quality. This explains our finding of null effects on correctness.

The fact that overall consistency substantially increases under the restatement method is more difficult to interpret. Does this reflect better answer quality? Or, do participants simply better recognize near-duplicate questions and try to be internally consistent?[32] To distinguish between these possibilities, our supplementary experiment examines answer quality using correlated question pairs rather than near-duplicates.

[32]Since there is direct monetary incentive for consistency, participants who recognize near-duplicates may be consistent due to a preference for consistency (Falk and Zimmermann, 2011) or because they think this will make it easier to earn restatement payments.

Figure 3: *Heterogeneity by strategy used in Restatement arm*



Notes: 95% confidence intervals relative to *No Incentives* (control) group. Restatement strategy was only asked to *Restatement* participants using an open-ended text-response question: "What was your main strategy for making sure you could correctly restate your answers?" We used OpenAI's API to code participants' responses, noting whether they answered carefully, memorized, or followed some other strategy (participants could be coded as following multiple strategies, see Appendix Section D.1). In the *Restatement* arm, 315 participants answered carefully, and 150 memorized (57 did both). 210 participants are classified as "Other", of which 196 had no particular strategy. Participants can be in multiple columns, since they can follow multiple strategies. Outcomes and specifications are otherwise the same as in Table 1.

5.3 Supplementary experiment: effect of *Restatement-Points* on correlations between strongly related questions

We ran a supplementary experiment (October–December 2024) to determine whether the restatement method improves answer quality when participants cannot rely on recognizing near-duplicates. The experiment uses question pairs from Pew surveys that measure related beliefs and policy preferences on the same topic. These questions naturally correlate in both Pew data and our sample, allowing us to test whether Restatement strengthens these correlations by reducing measurement error. The design mirrors our main experiment with specific modifications outlined below.

5.3.1 Supplementary experiment design

Treatments. Given the evidence that *Restatement-Points* was effective at increasing consistency in the main experiment, we include two treatments: *Restatement-Points* (N=702) and *No Incentives* (N=706). Both groups are paid a £4 completion fee, and are not paid to correctly restate answers in the restatement stage, yielding a clean comparison in which total payments are always exactly the same. The instructions are otherwise the same as in the main experiment. The samples were balanced across treatments, with a p-value of 0.71 on the joint F-test of differences across groups (Table A.6).

Stages. The stage structure was the same as in the main experiment, except the spillover stage was removed.

Main stage questions and outcomes.

Correctness. To replicate the results on correctness without any concern about near-duplicate questions, we included 14 correctness questions. We started from the same set as the main experiment, but replaced questions that could be perceived as duplicates (e.g., one

of the two dot counting tasks) with additional Raven's matrices.[33]

Correlation. To measure answer quality on unverifiable questions that were *not* near-duplicates, we included 12 pairs of questions from Pew's American Trends Panel Survey (ATP) (Pew Research Center, 2024) that are highly correlated in the Pew data (average $|\rho|$=0.34), and are a priori strongly related (see Table A.7 for question details). In each pair, one question concerns a *belief* related to a policy issue, and another concerns a corresponding *policy preference*.[34] If *Restatement-Points* increased answer quality, we would expect to see less noise in the data and thus higher correlations within pairs.

Empirical strategy for correlation. For each pair, we call the first question in the survey *Answer 1*, and the second question *Answer 2*. We standardize both so that the standard deviation in the *No Incentives* group is 1.[35] Then, the specification for participant i and question-pair j is:

$$Answer1_{ij} = \alpha Answer2_{ij} + \beta RestatePoints_i + \delta(Answer2_{ij} \times RestatePoints_i) + \varepsilon_{ij} \quad (4)$$

where α denotes the average correlation between the two answers in the *No Incentives* group and δ is the main coefficient of interest that indicates whether the correlation changes in the *Restatement-Points* arm.

5.3.2 Supplementary experiment: results

The *Restatement-Points* treatment does not increase the correlation between these pairs (β=-0.025, p=0.16, Table 2, columns 3-4), despite our large sample size (N=1408) provid-

[33]Specifically, we removed questions 2, 4, 10, and 14 from Table A.1.

[34]For example, we asked participants (3a) "How satisfied are you with the way democracy is working in the U.S.?" (Options: *Very satisfied, Somewhat satisfied, Not too satisfied, Not at all satisfied*) and (3b) "How much would you say the political system in the U.S. allows people like you to have an influence on politics?" (Options: *A great deal, A fair amount, Not too much, Not at all*).

[35]All answer options are either ordered multiple choice (which we treat as integers) or binary. For pairs with negative correlations, we reverse-code one of the questions in the pair so that all correlations are expected to be positive in our data.

ing sufficient power to detect even small changes in correlation of 0.05.[36] We also use the supplementary experiment to replicate the results on correctness without potentially distracting near-duplicate questions, and again find no effect on correctness (p=0.48, Table 2, columns 1-2). Together, these results make clear that *Restatement* only enhances consistency on near-duplicate questions, and does not improve other measures of answer quality. This strengthens our conclusion from Section 5.2 that focusing on memorization undercuts average quality, but leaves open the possibility that refined methods could increase answer quality without inducing memorization.

Table 2: *Supplementary experiment: impact on correctness and correlations.*

	Correct Answer (=1)		Correlation pair: Answer 1	
	(1)	(2)	(3)	(4)
Restatement-Points	−0.012	−0.007	−0.005	−0.004
	(0.012)	(0.011)	(0.015)	(0.014)
Answer 2			0.471***	0.463***
			(0.013)	(0.012)
Answer 2 × Restatement-Points			−0.026	−0.025
			(0.018)	(0.018)
Num. observations	19 684	19 684	16 872	16 872
Num. participants	1408	1408	1408	1408
Dep. var. control mean	0.62	0.62	0.00	0.00
R-squared	0.11	0.15	0.22	0.23
Question FEs	X	X		
Question-pair FEs			X	X
LASSO controls		X		X

Notes: Data from the supplementary experiment. Columns 1-2 use the same specification as Table 1. Columns 3-4 specification is Equation 4. *Answer 1* is the first question in a pair in the survey, *Answer 2* is the second question in a pair in the survey. All correlation questions have ordered lists as responses, which are coded as integers, then standardized so that the mean is 0 and standard deviation is 1 in the *No Incentives* (control) group. ρ, the average correlation between the two options in a pair, is thus 0.46. Standard errors clustered at the participant level. Omitted category is the *No Incentives* arm.

[36] This assumes α=5%, β=80%, implying that minimum detectable effect is $2.8\hat{\sigma}$ where $\hat{\sigma}$ is the standard error of the estimate.

6 Conclusion

This paper tests two psychological mechanisms to improve survey response quality. The *Bonus* method provides unexpected mid-survey payments framed as gifts to encourage reciprocal effort. The *Restatement* method promises payments for correctly restating previous answers, aiming to motivate careful initial responses that respondents can naturally reconstruct later. Both methods can be applied to unverifiable questions, explained in natural language, and incorporated into surveys with minimal increases in survey length.

Bonus payments improve both response correctness and consistency by 2-3 percentage points. These improvements are greater than the effects of dropping participants who fail an attention check, stemming from increased respondent effort and reciprocity toward researchers, consistent with gift exchange effects in other settings. By contrast, the restatement method does not clearly improve answer quality despite increasing consistency by 8 percentage points on near-duplicate questions. It falls prey to a multitasking effort problem that undermines answer quality: some participants reallocate effort toward memorizing their answers instead of answering more carefully. However, the results are consistent with a subset of respondents increasing answer quality, suggesting that future research could explore refined methods that do not induce respondents to memorize. For example, informing participants that they will later have to *explain their reasoning process* behind a previous answer, and evaluate the coherence between that explanation and their initial response, could encourage careful responses while minimizing thoughtless memorization.

Our findings demonstrate both the promise and challenges of psychological incentives in surveys. Improving answer quality appears fundamentally difficult: conventional direct payments for correct answers fail, and the *Bonus* method yields modest effects. The *Bonus* method may have other limitations: it may exacerbate experimenter demand effects, and it is unclear what the equilibrium effects would be if it were to become widespread practice (e.g.,

if participants learn to expect a bonus payment midway through surveys). Nevertheless, the success of the *Bonus* method relative to conventional monetary incentives and attention checks makes it a promising direction for future work.

Future research could extend this work in several directions. First, it could explore the practicalities of implementing the *Bonus* method. Can it work for in-person surveys or repeated waves? How does it interact with other methods, and what is the optimal timing within a survey? Second, researchers could investigate how to amplify the bonus effect, e.g., by increasing researcher salience to enhance reciprocity, or by combining it with norms messaging and commitment devices like accuracy pledges (Jacquemet et al., 2019). Third, inspired by our *Restatement* results—and the fact that it does increase effort—researchers may develop new mechanisms to better channel this effort toward the goal of increasing response quality.

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Online Appendix

A Supplementary tables and figures

Figure A.1: *Summary of experimental design*

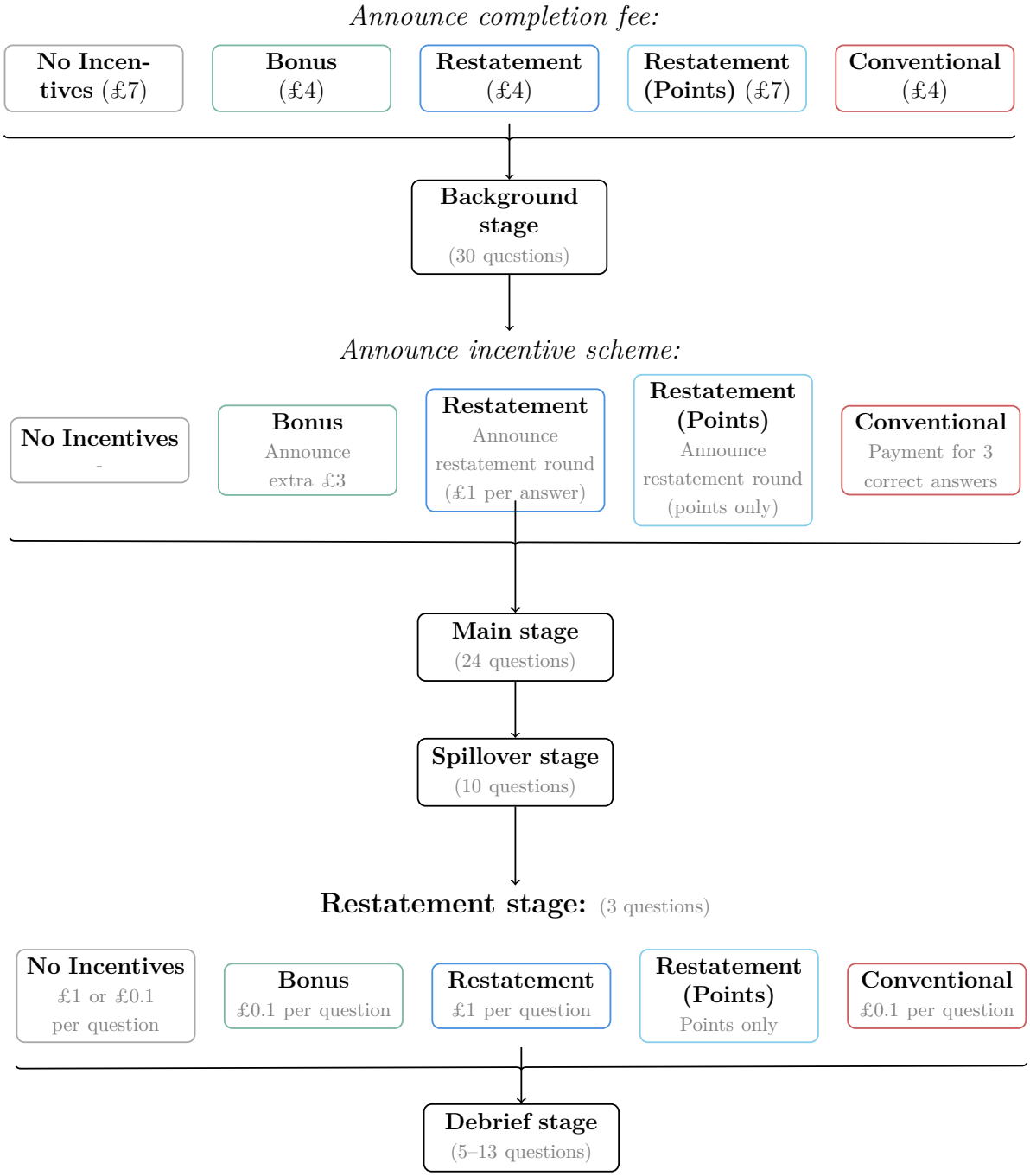

Table A.1: *Correctness questions in the main stage*

ID	Question wording
1	"A lottery ticket pays an amount of money that depends on chance. [*Probabilities are (45%, 35%, 20%), and one option first-order stochastically dominates.*] Which lottery do you most prefer?"
2	"A lottery ticket pays an amount of money that depends on chance. [*Probabilities are (40%, 35%, 25%), and one option first-order stochastically dominates.*] Which lottery do you most prefer?"
3	"William worked as a swim teacher for young kids for 5 weeks. In the first week, he worked 11 hours. In the second week, he worked 12 hours, and another 12 hours in the third week. The fourth week, he worked 8 hours. The fifth week, he worked 9 hours. How many hours did he work per week, on average?"
4	"For 5 weeks, Jennifer volunteered as a helper for swimming classes. The first week, she volunteered for 8 hours. She volunteered for 12 hours in the second week, and another 12 hours in the third week. The fourth week, she volunteered for 9 hours. The fifth week, she volunteered for 11 hours. For how many hours did she volunteer per week, on average?"
5	*Raven's matrix: dark arrows*
6	*Raven's matrix: arrows & circles*
7	"A bat and a ball cost $1.10 in total. The bat costs $1.00 more than the ball. How much does the ball cost?"
8	"If it takes 5 machines 5 minutes to make 5 widgets, how long would it take 100 machines to make 100 widgets?"
9	*Dot counting A (29 dots)*
10	*Dot counting B (31 dots)*
11	"The price of a pair of sneakers was $95 for the last six months of last year. On January 1st, there was a sale and the price decreased by 50%. 2 weeks later, the price increased by $5 and then Nina bought the sneakers at the new price. How much did Nina pay?"
12	"Dharik lives in a house on a straight street. For years, there have been 16 houses on his street to the right of his house and 17 houses on his street to the left of his house. Last year, 5 new houses were built on the same street even further to the left of those houses to the left of Dharik's house. If these are the only houses on this street, how many houses are on this street, including Dharik's house?"
13	*Reading comprehension (Mrs Smith)*
14	*Reading comprehension (house)*

Notes: All correctness questions used in the main stage. Also included were two Bayesian updating questions. These were pre-registered as exploratory, i.e., excluded from the main analysis. The questions were: (15) "The red jar contains 2 red balls and 1 blue ball. The blue jar contains 2 blue balls and 1 red ball. [*Description of 50% prior*] You reach your hand into the chosen jar, not knowing which jar it is. You have pulled out 1 blue ball. What is the probability that the red jar was chosen?" and (16) "The green jar contains 2 green balls and 1 purple ball. The purple jar contains 2 purple balls and 1 green ball. [*Description of 50% prior*] You reach your hand into the chosen jar, not knowing which jar it is. The first ball is green. The first ball is placed back in the jar. You then take another ball at random from the same jar. The second ball is also green. What is the probability that the green jar was chosen?"

Table A.2: *Consistency questions in main stage*

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ID	Question wording	"Verifiable" variant
1a	"Suppose that New York City still has a 'stop and frisk' policy in the year 2035. What percentage of people being stopped in 2035 do you think will be Black or Latinx?"	Ask about actual % in 2018
1b	"Suppose that New York City still has a 'stop and frisk' policy in the year 2035. What percentage of people being stopped in 2035 do you think will be NEITHER Black NOR Latinx?"	Ask about actual % in 2018
2a	"California and Texas are sometimes thought to be the two states with the most serious problem with unauthorized immigrants. Apart from these states, which of the following states do you think has the most serious problem with unauthorized immigrants?"	Ask about number of unauthorized immigrants instead of "most serious problem"
2b	"Apart from California and Texas, which of the following US states do you think has the biggest problem with unauthorized immigrants?"	Ask about number of unauthorized immigrants instead of "most serious problem"
3a	"In the year 2050, what proportion of total income in the U.S. do you think will be earned by the 20% of people who earn the most (before tax and redistribution)?"	Ask about actual proportion in 2023
3b	"In the year 2050, what proportion of total income in the U.S. do you think will be earned by the 80% of people who earn the least (before tax and redistribution)?"	Ask about actual proportion in 2023
4a	"Which of these countries do you feel values education (primary, secondary and post-secondary non-tertiary) the most?"	Ask about country that spends most on education instead of values it the most
4b	"In terms of the value it places on education (primary, secondary and post-secondary non-tertiary), which of these countries do you feel ranks the highest?"	Ask about country that spends most on education instead of values it the most

Notes: All consistency questions used in the main stage. (1a) and (1b) are a consistency pair, whose answers should add up to 100% to be internally consistent. (3a) and (3b) must also add up to 100%. (2a) and (2b) should yield same answers; (4a) and (4b) should also yield the same answers. 50% of participants in the *Restatement*, *No Incentives*, and *Bonus* treatments were randomized to receive the "Verifiable" variant; the others received the "Unverifiable" variant.

Table A.3: *Positive correlation between correctness and consistency*

	P(correct)		
	(1)	(2)	(3)
P(consistent)	0.326***	0.200***	0.268***
	(0.017)	(0.015)	(0.033)
P(consistent) × Bonus			−0.005
			(0.047)
P(consistent) × Conventional			0.084
			(0.054)
P(consistent) × Restatement			0.154***
			(0.047)
P(consistent) × Restatement-Points			0.197***
			(0.058)
Num. participants	2428	2428	2428
R-squared	0.14	0.35	0.16
Dep. var. control mean	0.74	0.74	0.74
LASSO controls		X	

Unit of observation is the participant i. $P(correct)$ is the probability that i got a question correct, i.e. the total number of correct answers divided by 14. $P(consistent)$ is the probability that i gave consistent answers, i.e. the total number of consistent answer-pairs divided by 4. LASSO controls include all the controls selected for both consistency and correctness in Table 1.

Table A.4: *Balance for main experiment*

	Means					p-values							
	(1)	(2)	(3)	(4)	(5)								
Variable	Control	Bonus	Conventional	Restatement	Rest. (Points)	(2)–(1)		(3)–(1)		(4)–(1)		(5)–(1)	
Female (=1)	0.49	0.50	0.42	0.49	0.47	0.57		0.06	*	0.99		0.58	
Age (years)	40.20	39.38	40.71	39.73	40.52	0.30		0.60		0.57		0.75	
Married (=1)	0.33	0.28	0.35	0.31	0.28	0.09	*	0.61		0.51		0.16	
Non-White (=1)	0.23	0.29	0.30	0.28	0.30	0.01	***	0.01	**	0.04	**	0.03	**
Completed College (=1)	0.58	0.57	0.64	0.61	0.63	0.80		0.07	*	0.35		0.17	
Mother Completed College (=1)	0.44	0.45	0.43	0.44	0.45	0.81		0.84		0.97		0.78	
Father Completed College (=1)	0.43	0.41	0.44	0.45	0.45	0.58		0.68		0.47		0.54	
Employed (=1)	0.66	0.66	0.68	0.65	0.62	0.93		0.56		0.74		0.26	
Passed Attention Check (=1)	0.90	0.86	0.91	0.89	0.90	0.03	**	0.55		0.64		0.88	
Born in USA (=1)	0.95	0.94	0.94	0.97	0.93	0.32		0.62		0.16		0.19	
Household Size (#)	2.68	2.76	2.70	2.66	2.74	0.33		0.85		0.78		0.54	
Children in Household (# under 18)	0.42	0.53	0.51	0.46	0.49	0.03	**	0.18		0.44		0.31	
Household Income ($1000s/year)	67.25	61.06	65.95	61.61	64.14	0.01	**	0.65		0.02	**	0.32	
Religious (=1)	0.43	0.43	0.45	0.41	0.44	0.90		0.61		0.42		0.67	
Homeowner (=1)	0.49	0.45	0.52	0.44	0.46	0.21		0.39		0.13		0.49	
Registered to Vote (=1)	0.92	0.90	0.89	0.88	0.90	0.34		0.26		0.07	*	0.53	
Altruism Index (Z-score)	0.00	-0.09	-0.01	-0.07	-0.05	0.11		0.89		0.27		0.52	
F-test: statistic						**1.63**		**1.12**		**1.06**		**0.63**	
F-test: p-value						**0.05**		**0.32**		**0.39**		**0.87**	

Notes: Columns (1)-(5) show the means of the demographic variables in each treatment group. Columns 6-9 show the p-value of the difference between each treatment group and the *No Incentives* control group. F-test is the test of the joint null of $\beta_1 = \beta_2 = ... = \beta_K = 0$ for the regression $Treat_i = \alpha + \sum_{k=1}^{K} X_i^k + \varepsilon_i$, where X_i^k are the demographic variables, $Treat_i$ is whether the individual is in the relevant treatment group or in the *No Incentives* control group.

Table A.5: *Attrition and selection*

Outcome var:	Completed survey (=1)		Predicted correctness			Predicted consistency		
Sample:	Started survey	Completed demographics	Completed demographics	Completed survey	Completed survey (adjusting for attrition)	Completed demographics	Completed survey	Completed survey (adjusting for attrition)
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Bonus	0.005	0.006	−0.013**	−0.014**	−0.014**	−0.009*	−0.010*	−0.010*
	(0.009)	(0.006)	(0.006)	(0.006)	(0.006)	(0.005)	(0.005)	(0.005)
Conventional	0.005	0.005	−0.003	−0.003	−0.003	−0.007	−0.007	−0.007
	(0.011)	(0.007)	(0.006)	(0.006)	(0.006)	(0.006)	(0.006)	(0.006)
Restatement	−0.016	0.001	−0.003	−0.004	−0.004	−0.008*	−0.009*	−0.009*
	(0.011)	(0.007)	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)
Restatement-Points	−0.017	−0.015	0.002	0.000	0.000	−0.006	−0.007	−0.007
	(0.014)	(0.011)	(0.006)	(0.006)	(0.007)	(0.006)	(0.006)	(0.006)
Num. participants	2507	2462	2462	2428	2428	2462	2428	2428
R-squared	0.00	0.00	0.00	0.00	0.00	0.00	0.01	0.01
Dep. var. control mean	0.97	0.99	0.75	0.75	0.75	0.60	0.60	0.60

Notes: Outcome in columns 1-2 is a binary indicator of whether the participant completed the survey. Column 1 includes all individuals who started the survey; column 2 includes all individuals who completed the baseline demographic survey. Outcome in columns 3-5 is *Predicted correctness*. This is constructed by predicting the correctness of each individual participant based on their baseline demographics (variables in Table A.4), using coefficients from an OLS regression of the baseline demographics on correctness score in the *No Incentives* group. Outcome in columns 6-8 is *Predicted consistency*, which is constructed similarly. Columns 3 and 6 start include all participants who completed the demographic survey; columns 4 and 7 include all participants who completed the survey (and therefore are included in the main specification). Columns 5 and 8 adjust for within-survey attrition by predicting the probability of attrition using a logistic regression with baseline demographics and reweighting the sample using inverse-probability weights. Standard errors in columns 3-8 are bootstrapped with 500 repetitions to account for error in the calculation of the inverse-probability weights and the predicted correctness and consistency scores.

Table A.6: *Balance for supplementary experiment*

	Means		p-values	
	(1)	(2)		
Variable	Control	Rest. (Points)	(2)–(1)	
Female (=1)	0.58	0.60	0.50	
Married (=1)	0.37	0.42	0.08	*
Non-White (=1)	0.36	0.39	0.20	
Completed College (=1)	0.63	0.64	0.79	
Mother Completed College (=1)	0.44	0.46	0.57	
Father Completed College (=1)	0.45	0.46	0.72	
Employed (=1)	0.67	0.68	0.65	
Born in USA (=1)	0.93	0.91	0.30	
Household Size (#)	2.98	3.10	0.14	
Children in Household (# under 18)	0.68	0.76	0.16	
Household Income ($1000s/year)	67.97	71.92	0.09	*
F-test: statistic			**0.73**	
F-test: p-value			**0.71**	

Notes: Columns (1)-(5) show the means of the demographic variables in each treatment group. Columns 6-9 show the p-value of the difference between each treatment group and the *No Incentives* control group. F-test is the test of the joint null of $\beta_1 = \beta_2 = ... = \beta_K = 0$ for the regression $Treat_i = \alpha + \sum_{k=1}^{K} X_i^k + \varepsilon_i$, where X_i^k are the demographic variables, $Treat_i$ is whether the individual is in the relevant treatment group or in the control group.

Table A.7: *Correlation questions in the supplementary experiment*

ID	Pew ID	Question subject	ATP Version	Correlation in Pew
1a	policy3	Scientific experts good at policy decisions	114	-0.46
1b	policy1b	Scientists have too much/too little influence		
2a	abrtlgl	Do you think abortion should be legal	110	-0.68
2b	abrtn_scotus	Do you approve of repeal of Roe v Wade		
3a	satisfied_democracy	Satisfied with democracy	105	0.34
3b	efficacy	Political system gives people like me a voice		
4a	intthreat_climatechange	How much of a threat, if at all, is climate change to our country?	105	0.73
4b	citizen_environment	How important is helping reduce climate change		
5a	smimpact_informed	Social media - made people more or less informed	105	0.31
5b	smeffect_awareness	Social media - effective for public awareness		
6a	fav_nato	Favorable about NATO	105	0.54
6b	benefitorg_nato	US benefits from NATO		
7a	transgend7	Discrimination against trans	109	0.46
7b	transgend6_a	Support protection against trans discrimination		
8a	techinfl	Tech have too much power/influence	88	0.38
8b	tc5	Tech should be regulated		
9a	ewtfreq	Extreme weather more than in the past	95	0.34
9b	infrgoal_d	Gov should set stricter standards for extreme weather		
10a	natprobs_k	Climate change is a problem	53	0.69
10b	clim1a	Climate is getting warmer because of people		
11a	natprobs_j	Economic inequality is a problem	53	-0.27
11b	taxsysm	Federal tax system is fair		
12a	natprobs_i	Illegal immigration is problem	53	0.65
12b	immva_lb	Should increase deportations		

Notes: All correlation questions used in the supplementary experiment. All questions taken from Pew's American Trends Panel Survey (Pew Research Center, 2024). *Pew ID* indicates the ID of the question in the original Pew dataset; ATP version indicates the wave of the ATP survey it was contained in. *Correlation in Pew* indicates the Pearson correlation between the two questions in the pair in the original Pew dataset, based on authors' own calculations. All questions had ordered lists as responses (e.g. (3a): "How satisfied are you with the way democracy is working in the U.S.?", options: *Very satisfied*, *Somewhat satisfied*, *Not too satisfied*, *Not at all satisfied*). These were coded as integers to calculate correlations. For pairs with *negative* correlations, one item was reverse coded in the supplementary experiement so that all correlations were in the same direction in the experimental data.

Table A.8: *Impacts are robust to multiple-hypothesis testing*

	Correct Answer (=1)		Consistent Answer (=1)	
	(1)	(2)	(3)	(4)
Bonus	0.006	0.021**	0.026	0.030**
	(0.012)	(0.010)	(0.014)	(0.013)
	[0.440]	[0.050]	[0.140]	[0.050]
Restatement	−0.020	−0.014	0.086***	0.084***
	(0.013)	(0.011)	(0.014)	(0.013)
	[0.200]	[0.150]	[<0.001]	[<0.001]
Restatement-Points	−0.020	−0.019	0.054***	0.070***
	(0.016)	(0.013)	(0.017)	(0.017)
	[0.230]	[0.140]	[0.010]	[<0.001]
Conventional	0.011	0.015	0.000	0.012
	(0.014)	(0.013)	(0.018)	(0.018)
	[0.410]	[0.170]	[0.600]	[0.310]
Num. observations	33 992	33 992	9712	9712
Num. participants	2428	2428	2428	2428
Dep. var. control mean	0.74	0.74	0.61	0.61
R-squared	0.07	0.14	0.22	0.25
Question FEs	X	X	X	X
LASSO controls		X		X

Notes: Uses q-values that are adjusted for multiple hypothesis testing to control for the false discovery rate Anderson (2008). Standard errors in parentheses, q-values in brackets. * $q<0.1$, ** $q<0.05$, *** $q<0.01$. Specification is otherwise same as Table 1. Columns (1) and (3) are treated as a multiple hypothesis testing family; columns (2) and (4) are treated as a family.

Table A.9: *Impacts on correctness and consistency are robust to alternative specifications*

	Correctness			Consistency		
	Correct Answer (=1) (RI p-values)	Correctness, Anderson index (Z)	Correct Answer (=1) Drop failed attention checks	Consistent Answer (=1) (RI p-values)	Consistency, Anderson index (Z)	Consistent Answer (=1) Drop failed attention checks
	(1)	(2)	(3)	(4)	(5)	(6)
Bonus	0.021**	0.116**	0.019*	0.030**	0.149***	0.032**
	(0.010) [0.030]	(0.050) [0.019]	(0.011) [0.076]	(0.013) [0.024]	(0.051) [0.003]	(0.014) [0.021]
Conventional	0.015	0.074	0.015	0.012	0.030	0.013
	(0.013) [0.257]	(0.065) [0.255]	(0.014) [0.279]	(0.018) [0.520]	(0.067) [0.652]	(0.019) [0.503]
Restatement	−0.014	−0.043	−0.019	0.084***	0.391***	0.087***
	(0.011) [0.191]	(0.050) [0.393]	(0.012) [0.106]	(0.013) [<0.001]	(0.051) [<0.001]	(0.014) [<0.001]
Restatement-Points	−0.019	−0.081	−0.013	0.070***	0.335***	0.070***
	(0.013) [0.149]	(0.065) [0.214]	(0.015) [0.395]	(0.017) [<0.001]	(0.067) [<0.001]	(0.018) [<0.001]
Num. observations	33 992	2428	30 212	9712	2428	8632
Num. participants	2428	2428	2158	2428	2428	2158
Dep. var. control mean	0.74	0.00	0.74	0.61	0.00	0.61
R-squared	0.14	0.28	0.12	0.25	0.12	0.25
Question FEs	X		X	X		X
LASSO controls	X	X	X	X	X	X
p-val: Bonus=Conv.	0.61	0.56	0.79	0.31	0.09*	0.31
p-val: Restate.=Bonus	0.00***	0.00***	0.00***	0.00***	0.00***	0.00***
p-val: Restate.=Conv.	0.03**	0.07*	0.02**	0.00***	0.00***	0.00***
p-val: Restate.=Restate-Points	0.69	0.59	0.70	0.38	0.48	0.32
p-val: Restate. (Mon. + Points pooled)=0	0.11	0.19	0.11	0.00***	0.00***	0.00***

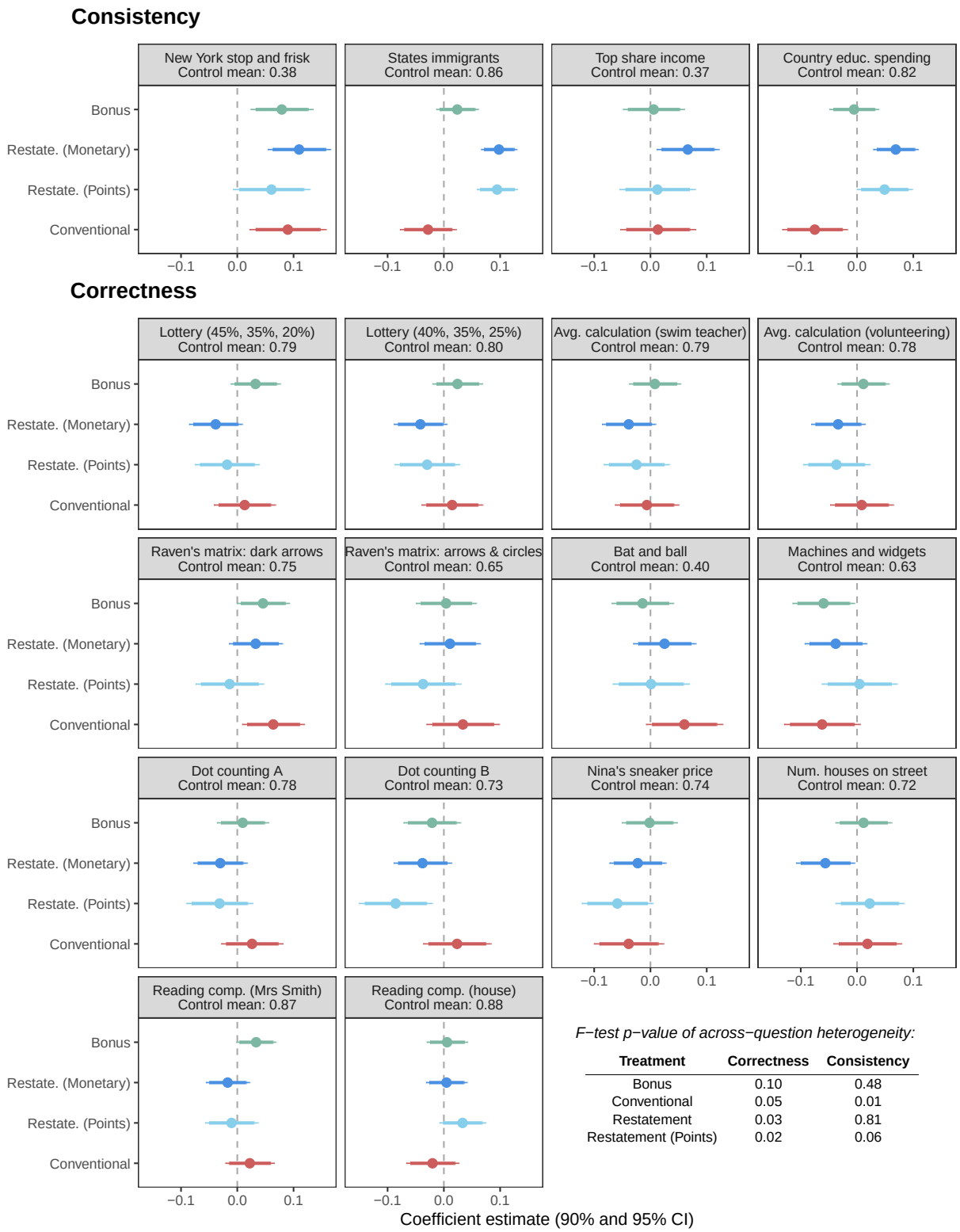
Notes: Standard errors in parentheses, p-values in brackets. Randomization inference p-values are used in columns 1 and 4, based on 500 simulations, otherwise specification is the same as Table 1. *Anderson index (Z)* are inverse-covariance-weighted indexes of at the participant level (Anderson, 2008). Columns 3 and 6 are the same as Table 1 but they drop the 270 participants who failed the attention check in the baseline stage.

Table A.10: *Bonus has larger impacts for easier questions*

	Correct (=1)
	(1)
Bonus	0.002
	(0.013)
Conventional	0.009
	(0.017)
Restatement	−0.014
	(0.014)
Restatement-Points	−0.025
	(0.017)
Above med. control mean (=1)	0.163***
	(0.009)
Bonus × Above med. control mean (=1)	0.035***
	(0.013)
Conventional × Above med. control mean (=1)	0.009
	(0.016)
Restatement × Above med. control mean (=1)	0.000
	(0.013)
Restatement-Points × Above med. control mean (=1)	0.009
	(0.016)
Num. observations	33 992
Num. participants	2428
Dep. var. control mean	0.74
R-squared	0.11

Notes: *Above med. control mean* indicates an "easy question": it is 1 when the mean proportion of correct answers for that question in the *No Incentives* group is above the median for all questions (0.748). Outcomes and specification are otherwise the same as Table 1.

Figure A.2: *Treatment effects by question*



F−test p−value of across−question heterogeneity:

Treatment	Correctness	Consistency
Bonus	0.10	0.48
Conventional	0.05	0.01
Restatement	0.03	0.81
Restatement (Points)	0.02	0.06

Notes: F-test p-value of across-question heterogeneity is from a regression that interacts the selected treatment with all question fixed effects, and then tests the null of all interaction terms being equal to 0.

Table A.11: *Effects are not specific to those with low or high baseline response time*

	Consistent Answer (=1)		Correct Answer (=1)	
	(1)	(2)	(3)	(4)
Bonus	0.030**	0.014	0.022**	0.018
	(0.013)	(0.019)	(0.010)	(0.014)
Conventional	0.012	0.024	0.015	0.027
	(0.018)	(0.026)	(0.013)	(0.018)
Restatement	0.084***	0.086***	−0.013	−0.014
	(0.013)	(0.018)	(0.011)	(0.015)
Restatement-Points	0.070***	0.073***	−0.019	−0.012
	(0.017)	(0.023)	(0.013)	(0.019)
Above med. baseline response time	−0.004	−0.008	0.017**	0.019
	(0.010)	(0.019)	(0.008)	(0.014)
Bonus × Above med. baseline response time		0.033		0.009
		(0.026)		(0.020)
Conventional × Above med. baseline response time		−0.022		−0.023
		(0.034)		(0.025)
Restatement × Above med. baseline response time		−0.004		0.001
		(0.026)		(0.021)
Restatement-Points × Above med. baseline response time		−0.005		−0.013
		(0.032)		(0.025)
Num. observations	9712	9712	33 992	33 992
Num. participants	2428	2428	2428	2428
Dep. var. control mean	0.61	0.61	0.74	0.74
R-squared	0.25	0.25	0.14	0.14
Question FEs	X	X	X	X
LASSO controls	X	X	X	X

Notes: Specifications are the same as in Table 1. *Above med. baseline response time* is 1 if the participant's average response time in the baseline demographics stage is above the sample median (15.8 seconds per question), and 0 otherwise.

Table A.12: *Consistency is slightly lower for verifiable questions, and is unaffected by question positioning*

	Consistent Answer (=1)			
	(1)	(2)	(3)	(4)
Bonus	0.030**	0.024	0.030**	0.030**
	(0.013)	(0.018)	(0.013)	(0.013)
Conventional	0.012	0.014	0.012	0.012
	(0.018)	(0.020)	(0.018)	(0.018)
Restatement	0.084***	0.084***	0.084***	0.084***
	(0.013)	(0.018)	(0.013)	(0.013)
Restatement-Points	0.070***	0.072***	0.070***	0.070***
	(0.017)	(0.018)	(0.017)	(0.017)
Verifiable questions (=1)	−0.021*	−0.025		
	(0.011)	(0.019)		
Bonus × Verifiable questions (=1)		0.012		
		(0.026)		
Restatement × Verifiable questions (=1)		0.001		
		(0.026)		
Position of pair (Z)			−0.008	
			(0.008)	
Bonus × Position of pair (Z)			0.015	
			(0.012)	
Conventional × Position of pair (Z)			0.003	
			(0.015)	
Restatement × Position of pair (Z)			0.002	
			(0.012)	
Restatement-Points × Position of pair (Z)			0.018	
			(0.014)	
Question distance (Z)				0.005
				(0.009)
Bonus × Question distance (Z)				0.002
				(0.012)
Conventional × Question distance (Z)				0.004
				(0.016)
Restatement × Question distance (Z)				−0.012
				(0.012)
Restatement-Points × Question distance (Z)				0.004
				(0.015)
Num. observations	9712	9712	9712	9712
Num. participants	2428	2428	2428	2428
Dep. var. control mean	0.61	0.61	0.61	0.61
R-squared	0.25	0.25	0.25	0.25
Question FEs	X	X	X	X

Conventional and *Restatement-Points* arms only used verifiable questions, so the interaction terms are excluded in column 2. *Position of pair (Z)* is order of appearance of the questions in the consistency pair, averaged over the two questions and then standardized to have a mean of 0 and standard deviation of 1. *Question distance (Z)* is a measure of the number of questions *between* the two questions in the consistency pair, standardized in the same way[55]Specification is otherwise the same as Table 1, but without LASSO controls.

Table A.13: *Impacts on correctness and consistency spill over to unincentivized round*

	Passed attention check (=1)	Correct Answer (=1)	Consistent Answer (=1)
	(1)	(2)	(3)
Bonus	0.003	0.029*	0.009
	(0.008)	(0.016)	(0.019)
Conventional	−0.005	0.029	0.031
	(0.011)	(0.021)	(0.025)
Restatement	0.005	0.014	0.078***
	(0.008)	(0.016)	(0.019)
Restatement-Points	0.015	0.002	0.121***
	(0.011)	(0.021)	(0.025)
Num. observations	2428	7284	2428
Num. participants	2428	2428	2428
Dep. var. control mean	0.98	0.71	0.83
R-squared	0.10	0.10	0.06
Question FEs		X	
LASSO controls	X	X	X
p-val: Bonus=Conv.	0.41	1.00	0.38
p-val: Restate.=Bonus	0.88	0.35	0.00***
p-val: Restate.=Conv.	0.35	0.47	0.05*
p-val: Restate.=Restate. (Points)	0.33	0.57	0.09*
p-val: Restate. (Mon. and Points pooled)=0	0.31	0.48	0.00***

In the spillover round, there are 4 correctness questions and 1 consistency pair. Columns 2 clusters standard errors at the participant level. Column 2 includes question fixed effects. In column 1, all LASSO controls selected for *both* consistency and correctness are included. The attention check question was asked at the end of the spillover round and was "*People are very busy these days and many do not have time to follow what goes on in the government. Some do pay attention to politics but do not read questions carefully. To show that you've read this much, please ignore the question below and select both "Extremely Interested" and "Slightly Interested". How interested are you in information about what's going on in government and politics?*"

Table A.14: *Comparison to attention check*

	Full Sample (1)	Passed Attention Check (2)	Effect Drop Failed Att. Check (2) - (1)	Bonus Effect	p-val. Bonus vs Drop
Panel A: All treatments					
Correctness	0.733	0.744	0.011	0.021	0.29
Consistency	0.643	0.650	0.007	0.030	0.10
N	2,428	2,158	-270 (11.1%)		
Panel B: No Incentives only					
Correctness	0.737	0.745	0.008	0.021	0.19
Consistency	0.608	0.614	0.005	0.030	0.06
N	609	548	-61 (10.0%)		

Notes: Shows the proportion of correct and consistent answers for the full sample (column 1) and when dropping people who fail the attention check (column 2). *Column 3*: the effect of dropping people who fail the attention check. *Column 4*: effect of *Bonus* from Table 1. *Column 5*: p-values comparing the effect of *Bonus* and the effect of dropping participants who failed the attention check. Null distribution to calculate p-values is calculated by randomly permuting the *No Incentives* and *Bonus* 1000 times, randomly permuting who failed attention checks 1000 times (keeping proportions fixed), and comparing the 2 values. *Panel A* includes all treatment groups for columns (1)-(3), *Panel B* includes only *No Incentives* participants. In Panel A, participants who *pass* the attention check are 6.4 pp (p<0.001) more correct and 4.8 pp (p=0.001) more consistent than those who *fail* the attention check.

Table A.15: *Prevalence of gaming, memorizing, and answering carefully*

Strategy	Treatment	% of Questions	% of Participants
Choosing easy-to-remember answers	Restatement	3.7%	14.5%
	Restate. (Points)	2.4%	11.0%
Searching for answers online	Restatement	0.8%	4.3%
	Restate. (Points)	0.5%	3.3%
Writing answers down	Restatement	2.4%	5.8%
	Restate. (Points)	1.3%	3.3%
Memorizing answers	Restatement	-	25.0%
	Restate. (Points)	-	26.3%
Answering carefully	Restatement	-	52.7%
	Restate. (Points)	-	49.0%

Notes: Participants were asked about the three gaming strategies explicitly, e.g. "Did you ever choose options that were easy to remember, even if they weren't necessarily the right answers, so that you could recall them better?" and then "For approximately what proportion of the questions in Section 2 did you choose options that were easy to remember, even if they weren't necessarily the right answers?". The *% of Questions* is the overall sample average of questions affected by the strategy, calculated by multiplying the proportion of participants who reported using the strategy for any question, and the average proportion of questions they report using the strategy for. *Memorizing answers* and *Answering carefully* were coded using OpenAI's API at the participant level from the text response to the question "What was your main strategy for making sure you could correctly restate your answers?" (see Appendix Section D.1 for more details).

Table A.16: *Results are similar when dropping participants who report gaming*

	Drop gamers		Effect of gaming	
	Correct (=1)	Consistent (=1)	Correct (=1)	Consistent (=1)
	(1)	(2)	(3)	(4)
Bonus	0.022**	0.031**	0.021**	0.030**
	(0.010)	(0.013)	(0.010)	(0.013)
Conventional	0.015	0.012	0.015	0.012
	(0.013)	(0.018)	(0.013)	(0.018)
Restatement	−0.009	0.081***	−0.010	0.081***
	(0.011)	(0.014)	(0.011)	(0.014)
Restatement-Points	−0.018	0.068***	−0.016	0.068***
	(0.014)	(0.018)	(0.014)	(0.017)
Gamer (Restate. arms only)			−0.019	0.014
			(0.017)	(0.018)
Num. observations	31 514	9004	33 992	9712
Num. participants	2251	2251	2428	2428
Dep. var. control mean	0.74	0.61	0.74	0.61
R-squared	0.14	0.25	0.14	0.25
Question FEs	X	X	X	X
LASSO controls	X	X	X	X
p-val: Bonus=Conv.	0.60	0.31	0.60	0.31
p-val: Restate.=Bonus	0.01***	0.00***	0.01***	0.00***
p-val: Restate.=Conv.	0.08*	0.00***	0.07*	0.00***
p-val: Restate.=Restate-Points	0.55	0.48	0.64	0.41
p-val: Restate. (Mon. + Points pooled)=0	0.23	0.00***	0.23	0.00***

Notes: Columns 1 and 2 drop all 177 participants in *Restatement* or *Restatement-Points* who report using any gaming strategy (choosing easy-to-remember answers, searching for answers online, writing answers down, see Table A.15). Columns 3 and 4 retain these participants but include a binary indicator for gamers as an explanatory variable.

Table A.17: *Participants in treatment groups were not more likely to select first or last options in the option list*

	First Answer	Last Answer
	(1)	(2)
Bonus	−0.002	−0.007
	(0.005)	(0.004)
Conventional	0.001	0.000
	(0.007)	(0.006)
Restatement	0.004	0.002
	(0.005)	(0.005)
Restatement-Points	0.003	0.008
	(0.006)	(0.006)
Num. observations	31 564	31 564
Num. participants	2428	2428
Dep. var. control mean	0.11	0.09
R-squared	0.03	0.03
Question FEs	X	X
LASSO controls	X	X

Notes: Only questions in the main stage with answer options (rather than text/numeric response) included. Unordered answer options were randomly reshuffled; ordered answer options were randomly selected to be "ascending" or "descending" (e.g., (1,2,3,4) or (4,3,2,1)). *First answer* is whether the respondent selected the first answer in the list of options as it appears to the respondent. *Last answer* is whether the respondent selected the last answer that appeared to the respondent.

B Proof

Proof of Proposition 1. Consider a participant who chooses effort levels $(e_1 \geq 0, e_2 \geq 0)$ to maximize the utility

$$U\big(e_1, e_2; B, R\big) \;=\; \underbrace{\gamma(B)\, p_Q\big(e_1, e_2\big)}_{\text{intrinsic payoff for accuracy}} \;+\; \underbrace{R\, p_R\big(e_1, e_2\big)}_{\text{restatement payoff}} \;-\; \underbrace{C\big(e_1, e_2\big)}_{\text{effort cost}}, \tag{5}$$

where

$$\begin{aligned} p_Q(e_1, e_2) \;&=\; p_0 \;+\; \alpha_1\, e_1 \;-\; \alpha_2\, e_2, \\ p_R(e_1, e_2) \;&=\; p_0' \;+\; \beta_1\, e_1 \;+\; \beta_2\, e_2, \\ C(e_1, e_2) \;&=\; c_1(e_1) \;+\; c_2(e_2). \end{aligned}$$

We assume $\alpha_1 > 0$, $\alpha_2 \geq 0$, $\beta_1 > 0$, $\beta_2 > 0$, and that $c_i(\cdot)$ are increasing and strictly convex, so $c_i''(\cdot) > 0$ and $c_i'(\cdot) > 0$.

First-Order Conditions. For a given B and R, the participant chooses (e_1, e_2) to maximize (5). The first-order conditions (FOCs) for an interior solution (e_1^*, e_2^*) are

$$\frac{\partial U}{\partial e_1} = \gamma(B)\, \alpha_1 \;+\; R\, \beta_1 \;-\; c_1'\big(e_1\big) \;=\; 0, \tag{6}$$

$$\frac{\partial U}{\partial e_2} = \gamma(B)\, \big(-\alpha_2\big) \;+\; R\, \beta_2 \;-\; c_2'\big(e_2\big) \;=\; 0. \tag{7}$$

Hence, at the optimum:

$$c_1'\big(e_1^*\big) = \gamma(B)\, \alpha_1 \;+\; R\, \beta_1, \tag{8}$$

$$c_2'\big(e_2^*\big) = -\,\gamma(B)\, \alpha_2 \;+\; R\, \beta_2. \tag{9}$$

Because c_1' and c_2' are strictly increasing (each c_i is strictly convex), these implicitly define unique effort choices $e_1^*(R)$, $e_2^*(R)$ for any given B and R. Inverse functions of the form $e_1^*(R) = \left[c_1'\right]^{-1}\big(\gamma(B)\,\alpha_1 + R\,\beta_1\big)$, $e_2^*(R) \;=\; \left[c_2'\right]^{-1}\big(-\,\gamma(B)\,\alpha_2 + R\,\beta_2\big)$ obtain if c_i' is invertible.

Condition for $\dfrac{\partial p_Q^*}{\partial R} > 0$. We define *initial accuracy*, or the probability of answering question Q correctly in the first stage, as $p_Q^*(R) \;\equiv\; p_Q\big(e_1^*(R),\, e_2^*(R)\big) \;=\; p_0 \;+\; \alpha_1\, e_1^*(R) \;-\; \alpha_2\, e_2^*(R)$. To study how $p_Q^*(R)$ changes with the restatement incentive R, take the total derivative:

$$\frac{d}{dR}\, p_Q^*(R) \;=\; \alpha_1\, \frac{\partial e_1^*}{\partial R} \;-\; \alpha_2\, \frac{\partial e_2^*}{\partial R}.$$

From (8) and (9), we have

$$\frac{\partial e_1^*}{\partial R} = \frac{\beta_1}{c_1''\big(e_1^*\big)}, \quad \frac{\partial e_2^*}{\partial R} = \frac{\beta_2}{c_2''\big(e_2^*\big)},$$

assuming differentiability of c_i' and $c_i'' > 0$. Therefore,

$$\frac{d}{dR}\, p_Q^*(R) = \alpha_1\, \frac{\beta_1}{c_1''\big(e_1^*\big)} \;-\; \alpha_2\, \frac{\beta_2}{c_2''\big(e_2^*\big)}. \tag{10}$$

Equation (10) reveals that raising R *increases* initial (Stage-1) accuracy p_Q^* if and only if

$$\alpha_1\, \frac{\beta_1}{c_1''\big(e_1^*\big)} \;>\; \alpha_2\, \frac{\beta_2}{c_2''\big(e_2^*\big)}. \tag{11}$$

In words, the term on the left captures how strongly the restatement incentive pulls up e_1—which helps accuracy through α_1—relative to how strongly the restatement incentive pulls up e_2—which *hurts* accuracy through α_2. The denominators $c_i''\big(e_i^*\big)$ represent the marginal cost of increasing each type of effort. If and only if (11) is satisfied, then $\frac{d}{dR}\, p_Q^*(R) \;>\; 0$

□

C Participant instructions

Figure A.3: *No incentives instructions*

Section 2

This section contains 24 questions.

We would like you to pay attention and put effort into answering the questions in this section.

Figure A.4: *Conventional Incentives instructions*

Section 2

This section contains 24 questions.

There are two types of questions:

- **Type 1**: questions with objectively correct answers
- **Type 2**: questions that ask you to select the option you most prefer

For these questions, you may receive additional payments based on the following procedure:

At the end of the experiment, we will randomly choose **three questions** from this section and evaluate your answers.

- For type 1 questions: you will receive **£1** for each correct answer
- For type 2 questions: you will receive the option that you choose

Because every question has a chance of being selected, it is in your best interest to simply choose the answer **you think is correct** (for type 1) or choose the option that **you most prefer** (for type 2).

We use this procedure because we would like you to pay attention and put effort into answering the questions in this section.

Figure A.5: *Bonus instructions*

Section 2

This section contains 24 questions.

To encourage you to answer the questions in this section carefully, we will provide you an extra bonus payment of **£3**. Hence your total payment will be **£7** if you complete the entire experiment.

We are providing this additional bonus because we would like you to pay attention and put effort into answering the questions in this section.

Figure A.6: *Restatement instructions*

Section 2

This section contains 24 questions.

For these questions, you may receive additional payments based on the following procedure:

At the end of the experiment, we will randomly choose **three questions** from this section and ask you to tell us the **same answers you gave previously**.

You will earn an extra **£1** for each of the same answers you provide, so that your additional payment can be up to **£3**.

The questions in this section ask for your personal opinion or involve solving simple problems. Your opinions are unlikely to change over the course of the survey, and you can always get the same answer on a problem by correctly solving it again. Therefore, by simply **answering the questions in this section carefully,** it should be easy for you to **give the same answers you gave previously**, which will **earn you more money.**

To encourage you to make an active choice on each question, the answer options on the randomly selected questions may be worded slightly differently and their order may be shuffled.

We use this procedure because we would like you to pay attention and put effort into answering the questions in this section.

Figure A.7: *Restatement-Points instructions*

Section 2

This section contains 24 questions.

For these questions, you will receive feedback about the quality of your responses based on the following procedure:

At the end of the experiment, we will randomly choose **three questions** from this section and ask you to tell us the **same answers you gave previously**.

You will earn an extra **1 point** for each of the same answers you provide, so that you can earn up to **3 points**. These points have **no monetary value**, but indicate high-quality responses.

The questions in this section ask for your personal opinion or involve solving simple problems. Your opinions are unlikely to change over the course of the survey, and you can always get the same answer on a problem by correctly solving it again. Therefore, by simply **answering the questions in this section carefully**, it should be easy for you to **give the same answers you gave previously**, which will **earn you more points**.

To encourage you to make an active choice on each question, the answer options on the randomly selected questions may be worded slightly differently and their order may be shuffled.

We use this procedure because we would like you to pay attention and put effort into answering the questions in this section.

D Data and measurement

D.1 Recall strategy coding and prompt

Model: GPT-4o-mini, temperature set at 0. The prompt was crafted based on best prompt-engineering practices and using Anthropic's "prompt improver".

```
You are an AI assistant helping researchers analyze responses from an online experiment. Your
    task is to categorize participants' strategies based on their responses to a question about
     how knowing they would have to restate their answers later affected their initial
    responses.

Here is the participant's response:

<participant_response>
{{RESPONSE INSERTED HERE}}
</participant_response>

Your goal is to determine whether the participant used any of the following strategies:

1. Extra attention: They answered with extra attention, effort, or truthfulness in their
    initial answers, taking care to provide accurate responses in the first place so they could
     restate more easily later.
2. Memorize: They tried to memorize their answers, so they could restate them later more easily
    .
3. Game the system: They tried to game the system (e.g., by writing down their answers,
    searching online, using AI, or other methods).
4. No strategy: They had no particular strategy.
5. Other: They used a strategy not covered by the above categories.

Please follow these steps:

1. Carefully read and analyze the participant's response.
2. In <strategy_analysis> tags, for each strategy:
   a. Provide arguments for why the participant might have used this strategy.
   b. Provide arguments for why the participant might not have used this strategy.
   c. Make a final decision on whether the participant used this strategy, based on the
       arguments.
3. Based on your analysis, create a JSON object with the following keys: extra_attention,
    memorize, game, no_strategy, and other. Assign a value of 1 to each strategy the
    participant used and 0 to those they didn't use.

Example output structure (do not copy the content, only the structure):

<strategy_analysis>
```

```
Extra attention:
Arguments for: [Reasoning]
Arguments against: [Reasoning]
Decision: [Used/Not used]

Memorize:
 [Same structure as above]

Game the system:
 [Same structure as above]

No strategy:
 [Same structure as above]

Other:
 [Same structure as above]
</strategy_analysis>

{
  "extra_attention": 0,
  "memorize": 0,
  "game": 0,
  "no_strategy": 0,
  "other": 0
}
```

D.2 LASSO controls

Following Belloni et al. (2014), we use double LASSO to select controls in the main results. The variables selected for the main specifications were:

Consistency (set $\mathcal{C}_1$): *Widowed (=1), Lives in Puerto Rico (=1), Lives in South Dakota (=1), # home purchases in last 12 months, Purchased Amazon Prime membership (=1), Purchased books (=1), Gets news from smart speaker, Gets news from smartwatch, Confidence in religious leaders, Confidence in business leaders, Response time (question 21), Verifiable or unverifiable.*

Correctness (set $\mathcal{C}_2$): selected variables include all those included in the consistency measures, plus: *African-American (=1), Asian-American (=1), High school degree (=1),*

Atheist (=1), Passed baseline attention check (=1), # home appliances owned, # modes of transportation used, Owns dishwasher (=1), Owns smoke alarm (=1), Owns vacuum cleaner (=1), Walking for commute (=1), Purchased spectator sports tickets (=1), Purchased DVDs or video streaming (=1), Got news from NYT (=1), Speaks Mandarin fluently (=1), Believes ABC News does own reporting (=1), Online access makes gov. more accessible (=1), Response time (question 10).

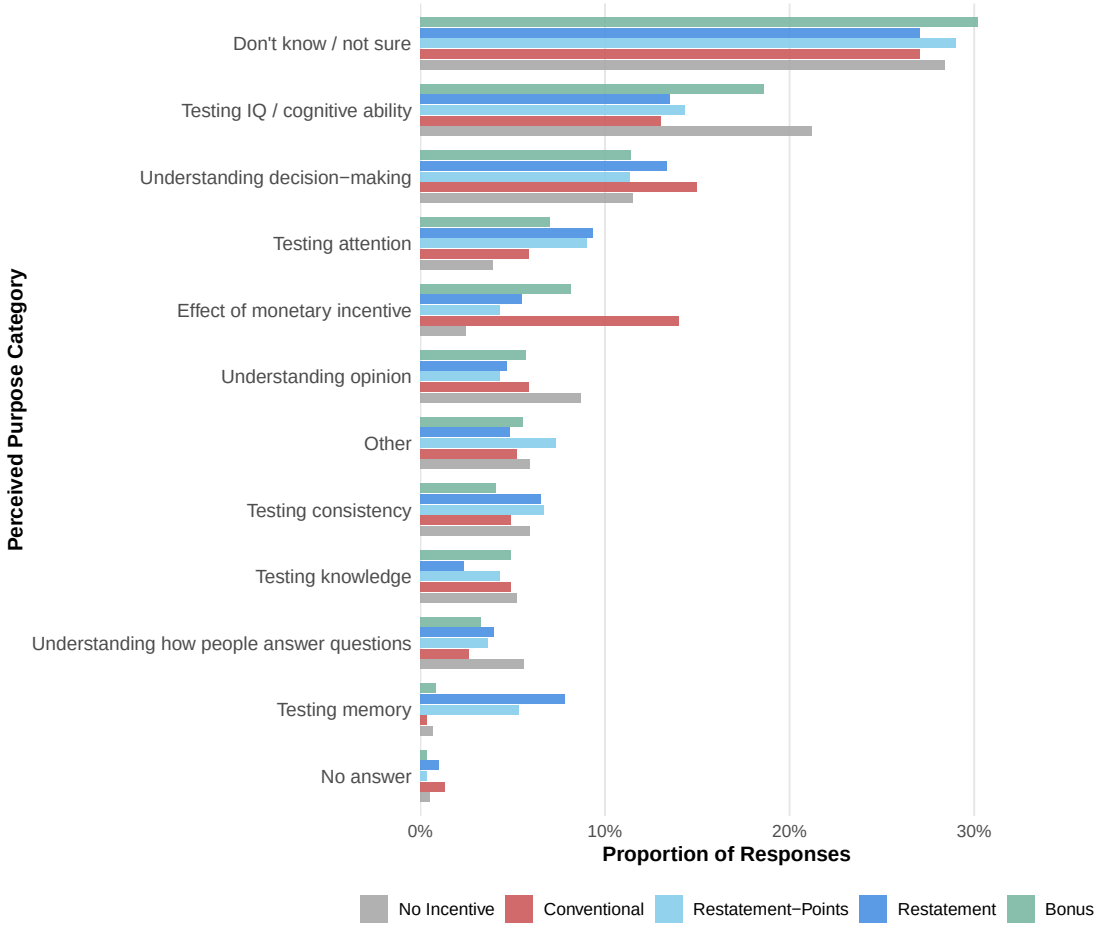
For other analyses, we control for (i) $\mathcal{C}_1$ when using consistency as an outcome; (ii) $\mathcal{C}_2$ when using correctness as an outcome; and (iii) the union of both sets ($\mathcal{C}_1 \cup \mathcal{C}_2$) when using other outcomes.

Table A.18: *Effects are robust to controlling for participants' perception of the purpose of the experiment*

	Consistent (=1)	Correct (=1)
	(1)	(2)
Bonus	0.029**	0.018*
	(0.013)	(0.010)
Restatement	0.086***	−0.014
	(0.013)	(0.011)
Restatement-Points	0.071***	−0.020
	(0.017)	(0.013)
Conventional	0.010	0.009
	(0.018)	(0.013)
Num. observations	9712	33 992
Num. participants	2428	2428
Dep. var. control mean	0.61	0.74
R-squared	0.25	0.15
Question FEs	X	X
LASSO controls	X	X
Perceived purpose controls	X	X
p-val: Bonus=Conv.	0.30	0.49
p-val: Restate.=Bonus	0.00***	0.00***
p-val: Restate.=Conv.	0.00***	0.08*
p-val: Restate.=Restate-Points	0.38	0.71
p-val: Restate. (Mon. + Points pooled)=0	0.00***	0.10

Notes: Both columns are main specifications but controlling for all possible categories of perceived purpose from Figure A.8 as dummies.

Figure A.8: *Perceived purpose of experiment by treatment group.*



Notes: Participants were asked what they thought was the purpose of the experiment in the final stage using a free-response text field. Answers were categorized using k-means clustering of OpenAI's semantic embeddings.